GRADUATION PROJECT / ANALYSIS & PLANNING

SmartListing Copilot

Multimodal Listing Quality Control for Fashion E-Commerce

MEMBER 1 | PROJECT PLANNING DOCUMENT

Document status: Revised-scope aligned planning baseline

Submission date: 16 October 2026

Authoritative basis: SmartListing Copilot - Project Proposal (Revised Scope), dated 12 October 2026

Scope assurance

Fashion Product Images Small is the sole MVP dataset. The planned system examines **five fashion product categories** (chosen after EDA) and the attributes **Product Type** and **Color** only. All suggested changes require human review; no automatic publishing is included.



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Document basis and interpretation

Evidence labels. Confirmed means prescribed in the Revised Scope proposal [1]; **proposed** means an implementation plan not yet demonstrated; **assumption** means a condition requiring later verification. The document does not represent interviews, tested performance, implementation completion or university approval.

Source hierarchy. Project-specific choices follow the Revised Scope [1]. The supplied Analysis & Design Blueprint [2] informs the analytical structure only; its unrelated example, numerical evidence, and engineering assumptions have not been adopted. No separately authored Member 1 draft or university cover-page template was supplied.

University milestones: 16 October (planning / literature / requirements), 6 November (analysis and design), 30 November (implementation), and 4 December 2026 (testing / final presentation).

1. Project Overview

SmartListing Copilot is a **proposed, pre-publication decision-support system** for the quality review of fashion e-commerce listings. It checks whether an uploaded product image agrees with the claims made in the product title and description. The intended users are marketplace sellers and catalog reviewers; operations and customer-support personnel are indirect beneficiaries. The project addresses possible contradictions in **Product Type** and **Color**, rather than attempting to validate an entire commercial catalog [1].

Two analytical paths operate in parallel: computer vision (CV) estimates the product type and color visible in the image, while rule-based natural language processing (NLP) extracts corresponding text claims. A comparison stage assesses contradictions, returns explainable reason codes and severity, and prioritizes a Review Queue. Controlled templates can propose a corrected title or description; the seller alone selects **Accept, Edit, or Ignore**. The FastAPI backend and Streamlit frontend support this human-controlled workflow [1].

The **expected value** is more consistent first-pass checking, focused human attention and actionable explanations before publication. These are planned benefits, not demonstrated reductions in costs, returns, complaints or review time. The prototype complements seller judgment; it does not autonomously approve or publish listings.

2. Problem Statement

2.1 Current situation

Fashion product information may be assembled and reviewed manually. Images, titles and descriptions can contain competing or incomplete representations of the same item. A reviewer may therefore need to compare these elements individually to decide whether the listing is internally consistent. The Revised Scope describes this operational problem, but provides no verified local time-study or marketplace incident data [1].

2.2 Identified problem

A listing may depict one product type while naming another, describe a color that differs from the image, or omit a relevant type or color claim. Such defects can make listing quality difficult to assess consistently and may create avoidable ambiguity for sellers and buyers. Manual identification of these conflicts can be repetitive, and a generic warning without evidence offers limited guidance about what should be corrected.

2.3 Desired situation and planning rationale

The desired state is an **explainable pre-publication check** that flags possible image-text disagreements, distinguishes critical issues from uncertain cases, and proposes bounded text corrections while retaining seller approval. Detecting inconsistency is valuable because it shifts attention to questionable listings and makes the basis of a suggested edit visible. It does not establish that the proposed system will reduce returns, expense or review duration; such effects require separate real-world validation.

3. Needs Assessment

This assessment translates the business problem described in the Revised Scope [1] into stakeholder needs and proposed responses. The needs are **inferred from the proposal**, not collected through interviews, surveys or direct field observation. No legal or industry compliance requirements are asserted.

Business problem	Stakeholder need	Proposed system response	Expected benefit
Image-text inconsistency	Identify contradictions before a listing is published	Compare the CV attribute prediction with claims extracted from title and description	Earlier visibility of questionable listing content
Incorrect product type	Know whether the named product type fits the image	Predict articleType-level product type within five chosen classes; flag TYPE_MISMATCH	More consistent product-type review

Business problem	Stakeholder need	Proposed system response	Expected benefit
Incorrect color information	Distinguish material color conflicts from near-shade differences	Predict baseColour-aligned color; compare normalized color families and apply COLOR_MISMATCH	More informative handling of visual color disagreements
Manual listing review	Direct attention to listings needing human judgment	Review Queue: Ready to publish, Needs review and Critical mismatch	Prioritized checking instead of undifferentiated review
Unexplained warnings	See evidence and the reasons behind a flag	Show predicted versus claimed attributes, confidence, severity and reason codes	Reviewable, interpretable findings
Seller-controlled corrections	Retain ownership of content changes	Template-based Before/After suggestion with Accept / Edit / Ignore	Actionable feedback without autonomous changes

3.1 Classification of needs

Proposal-inferred felt needs: Seller clarity, reviewer prioritization, understandable warnings and control over corrections are reasonable potential user preferences, but have not been directly confirmed by stakeholders. **Normative or engineering-quality needs:** Traceability, bounded outputs, explicit uncertainty, and reproducible evaluation are proposed technical quality expectations derived from the project design, not cited regulatory obligations.

Validation later: Confirm terminology, warning comprehension, queue usefulness and editing behavior through scenario walkthroughs or usability assessment. This is a proposed validation activity; no acceptance feedback has been gathered.

4. Project Objectives

Overall objective. Design, implement and evaluate a limited-scope, human-controlled multimodal workflow that identifies and explains Product Type and Color inconsistencies between fashion product images and listing text using Fashion Product Images Small [1].

The objectives below are identifiable and measurable through deliverable inspection or future tests. No performance values are claimed or set before baseline development. **Validation thresholds will be fixed on validation data after baseline experiments and before held-out evaluation.** IDs are stable planning references for Members 5 and 6, not a replacement for their detailed requirements IDs.

Objective ID	Objective description	Expected deliverable	Measurement method
OBJ-01	Establish the end-to-end image + text listing-checking pipeline	Integrated prototype workflow	Scenario test from submission to final review decision
OBJ-02	Predict visual Product Type and Color for the selected five classes	Transfer-learning CV module and CLIP zero-shot baseline	Accuracy, macro-F1, confusion matrix; calibration / ECE
OBJ-03	Extract Product Type and Color claims from title / description text	English rules and bounded Arabic lexicon	Precision, recall and F1 by claim and language
OBJ-04	Detect discrepancies between CV outputs and NLP claims	Rule comparison plus lightweight mismatch classifier	Per-error precision, recall, F1 and false-alarm rate
OBJ-05	Attach understandable evidence and reason codes to every warning	Mismatch report with severity, explanation and confidence	Completeness checks for warnings and explanation mapping
OBJ-06	Prioritize listings for human review	Three-state Review Queue and inconsistency routing	Case-based routing checks, including low-confidence cases
OBJ-07	Generate bounded suggested title / description corrections	Type-and-color-only templates and Before/After view	Automatic attribute-traceability checks; scenario inspection

Objective ID	Objective description	Expected deliverable	Measurement method
OBJ-08	Ensure that suggested edits remain under seller control	Accept / Edit / Ignore workflow; no auto-publish behavior	Interaction tests confirm no suggestion is applied without a decision
OBJ-09	Evaluate the pipeline with transparent evidence and limitations	Evaluation protocol and results report (future deliverable)	Separate synthetic vs manually constructed metrics; catch rate and response time

Evaluation discipline: product-ID splitting precedes error injection; synthetic and manual-test results remain separate. Numerical acceptance thresholds are to be established and frozen on validation data, not retrofitted to held-out results. [1]

5. Stakeholder Analysis

The groups below are taken from, or directly implied by, the Revised Scope [1]. **Influence** is a preliminary estimate of ability to shape scope or acceptance, while **project impact** is the expected degree of effect on the group. Both classifications are planning judgments, **not empirically validated stakeholder findings**.

Stakeholder	Role	Interest	Expected needs	Influence	Project impact
Marketplace sellers	Primary decision makers	Clear listings before publication	Simple submission, explainable flags, controlled edits	High	High
Catalog reviewers	Primary review users	Consistent content triage	Prioritized queue, severity, evidence	Medium	High
Development team	Build and evaluation	Implementable, bounded scope	Defined dataset, stable contracts, test cases	High	High
Operations / customer support	Indirect beneficiaries	Fewer ambiguous listings	Consistent listing-quality information	Low	Medium
Academic supervisor (if applicable)	Governance and assessment	Scope clarity and defensible evidence	Well-defined deliverables, academic integrity	High*	Medium*

* Conditional: the supervisor role and its rating require confirmation with the academic setting. No interviews, sign-offs, deployment sponsors, administrator roles or user-account requirements are assumed.

6. Project Scope

6.1 In scope - approved MVP

The MVP is a fashion-only, pre-publication **listing-consistency** prototype for **exactly five** product types selected after EDA. Confirmed core capabilities are:

- Accept a single listing containing a product image, title, description and seller category.
- Run ResNet or MobileNet transfer learning for visual Product Type and Color; benchmark against CLIP zero-shot, which is a baseline and fallback rather than a preprocessing step.
- Perform rule-based English claim extraction with a small fixed Arabic vocabulary, then apply attribute-comparison rules and a lightweight classifier.
- Provide confidence-aware mismatch severity, reason codes and explanations; route listings to a three-state Review Queue.
- Offer controlled, type/color-grounded template suggestions with a Before/After view and mandatory Accept / Edit / Ignore seller action.
- Use FastAPI and Streamlit for the proposed backend and user-facing interface.

6.2 Simplified supporting capabilities and non-mandatory options

The Revised Scope [1] identifies limited supporting features: a transparent **Listing Quality Score** (image quality, attribute completeness, consistency), Product Type/Color **Missing Attribute Detection**, and basic visual-quality checks. An optional CLIP linear probe and basic MLflow/Docker support are not essential to the core workflow and may be deferred if necessary. The project plan does not expand these into new MVP requirements.

6.3 Out of scope

Return prediction; dynamic pricing; price validation or price sanity checks; material or size recognition; LLM-generated content; autonomous publishing; live marketplace integration; production cloud deployment; broad multilingual NLP; additional commerce domains; competitor scraping; recommendation models; customer-behavior prediction; full multimodal transformer development; batch CSV processing; and LoRA fine-tuning. These are excluded rather than silently treated as secondary deliverables.

6.4 Scope constraints and decision gates

- Only Fashion Product Images Small supplies images and underlying product metadata for MVP training, validation, testing and pipeline evaluation; constructed listings reuse those images rather than introduce another dataset.
- The final five articleType categories are **not yet selected**; EDA must assess class balance, image availability and label quality before the selection is frozen.
- Only **Product Type** and **Color** can be checked as core semantic attributes; Arabic support is deliberately restricted and seller verification is mandatory.
- No list of five example product types is treated as the approved final class set, and no claim of deployed integration is made.

7. Preliminary Feasibility Study

This is a **pre-implementation feasibility assessment**, not evidence that the proposed end-to-end system has been built or validated. It distinguishes technology availability from dataset- and project-specific feasibility, which still require experimentation [1, 3-6].

7.1 Technical feasibility

Dataset availability and fit. The publicly listed Fashion Product Images (Small) dataset contains fashion images and a styles.csv metadata map connected by product ID [3]. The proposal assigns **articleType** to product-type supervision, **baseColour** to color supervision and **productDisplayName** to title analysis [1]. This is a plausible supervised classification and consistency-checking basis, but class coverage, label ambiguity, image availability, file integrity and suitability of the final five categories remain unverified pending EDA.

Image models. ResNet and MobileNet are established families of image-classification architectures, and pretrained-weight workflows are available in standard ecosystems [4, 5]. Transfer learning is therefore *technically plausible*, not yet demonstrated to meet the project's target on this dataset. CLIP zero-shot [6] is proposed as an independent baseline and fallback, not a sequential prerequisite to the supervised model. Limited image resolution, background variation, similar shoe types, color ambiguity and unseen categories may constrain reliable predictions.

NLP and detection. Dictionary/regex-based English extraction and a bounded Arabic lexicon are feasible with standard Python tooling. Their coverage will be limited by ambiguous terms, spelling, dialect, Arabizi, stylized color names and incomplete descriptions. Comparing normalized claims to CV outputs by rules is implementable; whether adding logistic regression or a small MLP improves the rule baseline remains an empirical question because mismatch training cases will be synthetically constructed.

Integration and resources. FastAPI, Streamlit, Pydantic and ordinary Python ML libraries offer a viable application stack [1, 7, 8]. A unified input/output schema and early stubbed components are proposed to reduce integration friction. Memory capacity, CPU/GPU availability, training time, pretrained-weight download access, image-loading throughput, latency and

model version compatibility must be checked on the team's actual equipment. No hardware capability or computation budget has been verified.

7.2 Operational feasibility

The proposed process begins when a seller provides an image and text. The system returns explicit type/color evidence, uncertainty indications and a queue priority, followed by a bounded edit suggestion. This could fit a pre-publication review process because the seller retains the final decision and catalog reviewers can focus on flagged cases. Operational fit is a design hypothesis: warning overload, confusing severity distinctions, unclear low-confidence messages, unrealistic synthetic text and limited Arabic coverage could reduce practical usability.

Human review also mitigates, but does not eliminate, the possibility of incorrect model predictions. The project has **no completed user acceptance testing**; interface comprehension, time-to-review, and the usefulness of queue decisions require later scenario-based assessment.

7.3 Economic and resource feasibility

The planned stack relies substantially on open-source frameworks and a publicly listed dataset, which may reduce software-license barriers. It does **not** establish zero development cost or zero compute cost. Potential resource needs include dataset storage, model training and inference, operating-system compatibility, annotation/review time for manual cases, integration labor, troubleshooting, reproducibility packaging and demonstration setup. The size of these costs depends on local machines, software licenses applicable to reuse, workload and team availability; no budget, ROI, hourly rate or purchase price is asserted.

The academic schedule is tight relative to a multimodal build. Feasibility therefore depends on protecting the core pipeline and retaining the supporting-feature cut order in the approved proposal. A numeric cost estimate should be produced only after actual compute and effort requirements are observed.

7.4 Feasibility summary

Feasibility dimension	Supporting evidence	Main constraints	Preliminary assessment	Further validation needed
Technical	Dataset fields [1, 3]; proven model families [4-6]; Python integration stack	Low-res images; category overlap; label noise; runtime and model behavior unknown	Plausible, conditionally feasible	EDA; baselines; class and color metrics; compute checks; integration tests
Operational	Human-controlled review and bounded warnings are specified in proposal	No real seller study; usability and false alarms not established	Workflow is conceptually coherent	Scenario walkthroughs; queue testing; user-feedback study if accessible
Economic / resources	Public dataset listing and open-source stack	Undetermined time, compute, storage, annotation and packaging effort	Potentially practical for a scoped academic prototype	Team availability; actual resource inventory; measured workloads / costs

Preliminary conclusion: Available methods support a reasonable prototype plan, but adequate dataset quality, predictive utility, performance, operational fit and cost remain open verification tasks.

8. Dataset Description

Confirmed dataset selection: *Fashion Product Images (Small)*, published on Kaggle by Param Aggarwal [3] and designated as the **single source** for the MVP in the Revised Scope [1]. The source listing describes approximately 44,000 products, with image files identified by ID and descriptive fields in **styles.csv**. This is source documentation, **not an independently audited row count or completed EDA result**.

An image file such as **images/<id>.jpg** is linked to the record containing the matching **id**. Metadata supplies supervised labels and existing titles. Any product description used in this project will be **manually authored or synthetically generated** and clearly marked by origin; the selected metadata does not contain a dedicated description column. Price information required for price validation is not provided and is outside scope [1].

8.1 Relevant metadata fields

Column	Planned role	Planning caution
id	Relate metadata to the product image; establish product-ID data splits	Check non-null values, image existence, duplicates and split leakage
articleType	Product Type label and candidate five-class selection	Inspect class counts, noisy labels and confusion among near types
baseColour	Color label for visual model and claim comparison	Normalize aliases / color families; assess multi-color ambiguity
productDisplayName	Existing product title for rule-based NLP analysis	Titles are real dataset metadata; verify whether type/color are explicit
masterCategory	Broader hierarchy for EDA and scope filtering	Do not mistake master-category names for final articleType classes
subCategory	Intermediate taxonomy to help group related product types	Evaluate hierarchy quality; grouping decisions remain proposed

8.2 Planned preparation and evaluation integrity

- Audit file structure, IDs, missing metadata, unmatched images, corrupt samples, duplicates, distribution and label quality; EDA results must be reported later, not presumed here.
- Select exactly five usable articleType classes after EDA based on class balance, visual separability, image availability and label quality.
- Normalize Product Type/Color vocabularies and prepare image resizing, cleaning, quality checks and model-specific transforms without implying recovery of invisible detail.
- Split by product ID into train, validation and test **before** synthetic wrong-color / wrong-type error injection; check for duplicates across splits. The proposal provides an illustrative 70/15/15 split, not an approved final ratio.
- Construct an independent, labeled manual evaluation set of **100-150 planned EN/AR listings** from the test-split images; include near shades, missing claims, swapped descriptions, multiple colors and stylistic language. This target is proposed, not achieved.
- Keep synthetic and manually constructed results separate; freeze tuning thresholds using validation data and do not calibrate them on the held-out manual set.

Limitations. Small images, possibly noisy or ambiguous color labels, curated rather than marketplace-native listings, no real seller descriptions, no return outcomes and restricted attribute coverage limit external generalization. The planned evaluation can test internal consistency behavior but cannot prove marketplace operating gains.

9. Proposed Solution and High-Level Workflow

The proposed application receives one listing, then executes visual and textual analysis as **parallel branches**. The image branch applies model-appropriate preprocessing and predicts Product Type and Color with a ResNet/MobileNet transfer-learning model; **CLIP zero-shot is independently compared as a baseline and available fallback**. The title/description branch extracts English claims, with only a small predefined Arabic vocabulary. Both branches produce structured attributes and evidence for the comparison engine.

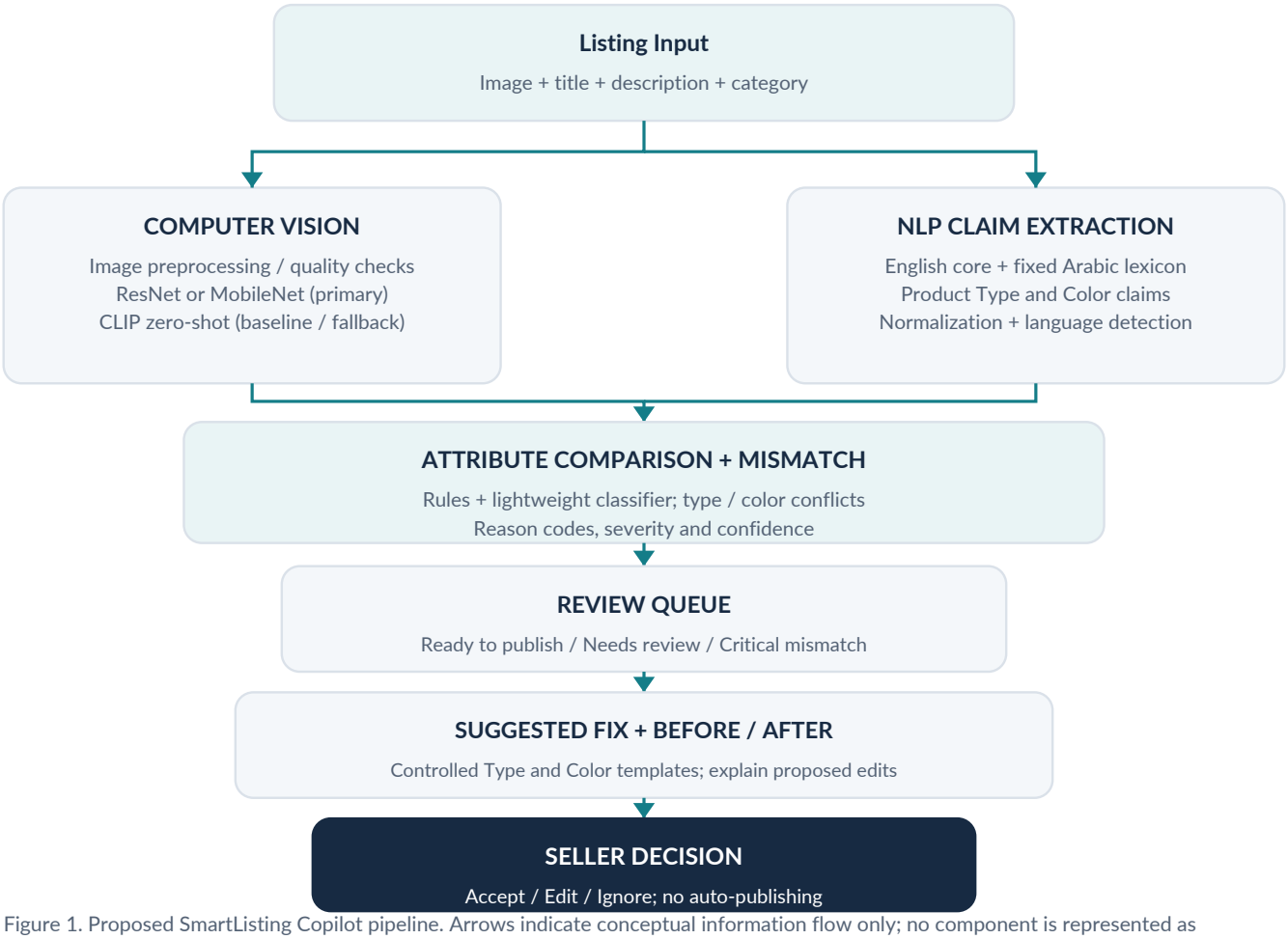


Figure 1. Proposed SmartListing Copilot pipeline. Arrows indicate conceptual information flow only; no component is represented as implemented or experimentally verified.

The comparison layer proposes rule-based severity levels and a lightweight classifier output, then attaches reason codes and explanations. The scoring and triage layer determines a Listing Inconsistency Level and Review Queue placement; image blur, unsupported categories and low confidence can trigger **Needs review** without implying a confident mismatch. Limited supporting capabilities include Missing Attribute flags and a three-component Listing Quality Score [1].

A controlled suggestion produces only model-supported Type/Color text, which is displayed beside the original as a Before/After view. **Template groundedness prevents invented attribute types; it does not guarantee a correct visual prediction.** The seller reviews and chooses Accept, Edit or Ignore. No listing is published automatically.

9.1 Explanation contract and uncertainty handling

Each reportable warning should identify the disputed or missing claim, predicted visual attribute, confidence when available, severity and plain-language rationale. Proposal-defined reason codes include **TYPE_MISMATCH, COLOR_MISMATCH, LOW_CONFIDENCE, IMAGE_BLUR, MISSING_ATTRIBUTE** and **UNSUPPORTED_CATEGORY** [1]. These are planning-level output concepts, not the detailed functional specification owned by Member 5.

Integration intent. A common Pydantic data contract can connect the CV, NLP, comparison, scoring and presentation modules; initial stub modules can be replaced incrementally. This is a proposed integration strategy rather than proof of a working system.

10. Expected Deliverables

10.1 Current planning submission - 16 October 2026

The official submission combines the team's **Project Planning, Literature Review and Requirements** materials. This document contributes **Member 1 Project Planning** only: overview, problem framing, inferred needs, objectives, stakeholders, scope, preliminary feasibility, dataset plan, conceptual workflow, success criteria and assumptions. Separate Member 2 scheduling, Member 3-4 reviews, Member 5 functional requirements and Member 6 non-functional requirements/use cases are outside this document. Their completion or approval is not claimed.

10.2 Future technical and evaluation outputs (planned)

Phase	Planned outcome
System Analysis & Design - 6 Nov 2026	EDA and finalized five classes; dataset schema/splits; proposed system design; model baseline plan
Implementation - 30 Nov 2026	Cleaned data pipeline; CV model and CLIP baseline; NLP claims; mismatch rules/classifier; queue and templates; integrated FastAPI + Streamlit prototype
Testing & Final Presentation - 4 Dec 2026	Held-out synthetic and manual evaluation report; confidence and error analysis; test evidence; reproducibility materials; demo and final documentation
Supporting deliverables (conditional)	Lightweight MLflow experiment logs, Docker package and optional CLIP linear probe only if core delivery remains secure

Possible reproducibility artifacts include requirements, README, notebooks, saved weights and API usage notes as envisioned by the proposal [1]. Detailed activity breakdowns and timelines belong to Member 2 and are not reproduced here.

11. Success Criteria

These are **proposed, testable MVP completion conditions**, not test results. Numerical model thresholds must be selected on validation data after baseline experiments and fixed before held-out evaluation [1].

Criterion	Testable completion condition
SC-01 End-to-end submission	A supported image + title + description passes through the UI and returns findings and a seller-decision interface.
SC-02 Attribute performance	Product Type and Color evaluation is reported separately using accuracy, macro-F1 and class/color breakdowns; approved thresholds are met or limitations/fallbacks are documented.
SC-03 Contradiction identification	Wrong-type, wrong-color, combined-error and correct-listing cases are tested with precision, recall, F1 and false alarms reported.
SC-04 Explainability	Every displayed warning maps to a defined reason code, evidence and understandable rationale.
SC-05 Review routing	Ready to publish, Needs review and Critical mismatch routing matches the agreed decision logic for test scenarios.
SC-06 Suggested corrections	Template output contains only supported Type/Color attributes; Before/After highlights proposed changes.
SC-07 Human control	Seller can Accept / Edit / Ignore; no change is published or applied autonomously.
SC-08 Uncertain cases	LOW_CONFIDENCE, IMAGE_BLUR and UNSUPPORTED_CATEGORY test cases are escalated appropriately; unsupported images are not confidently auto-accepted.

Criterion	Testable completion condition
SC-09 Evaluation separation	Metrics from synthetic injected-error cases and the separately constructed manual set are reported distinctly, not combined.
SC-10 Complete pipeline report	End-to-end catch rate and response time are measured; major limitations, threshold selection and data provenance are documented.

The manual test set is constructed from held-out dataset images rather than real marketplace submissions. Passing these criteria would demonstrate an academic proof of concept only, not marketplace deployment, commercial value or reduced return rates.

12. Planning Assumptions and Dependencies

12.1 Confirmed decisions versus open assumptions

Confirmed design decisions: Fashion Product Images Small as the sole dataset; exactly five classes chosen after EDA; Product Type and Color only; transfer-learning CV with CLIP zero-shot baseline/fallback; bounded English/Arabic rules; explainable review queue; controlled fixes; seller decision; FastAPI and Streamlit; four official milestone dates [1].

Assumptions needing confirmation: The released files and metadata can be joined reliably; at least five sufficiently represented types exist with usable labels; image quality permits meaningful type/color evaluation; team hardware can run training and inference; the CV/NLP/comparison contracts integrate without blockers; and the manually constructed evaluation set can be produced and quality-checked. No assumption above is presented as an accomplished result.

12.2 High-level dependency relationships

Predecessor or input	Dependent activity	Why the dependency matters
Dataset access + audit	EDA and five-class selection	Final class scope must follow actual images, class balance and label quality.
EDA + selected categories	Metadata cleaning and product-ID splits	Stable labels and leakage-safe partitions are prerequisites for valid experiments.
Splits + error-injection protocol	Mismatch model training and evaluation	Injected variations from the same product must not cross train/test boundaries.
CV output contract + NLP claim contract	Attribute comparison and explainability	Comparable normalized type/color fields and confidence markers are needed.
Comparison + reason codes	Queue routing and template suggestions	Findings supply severity, grounded attributes and review reasons.
Module interfaces + API integration	Streamlit end-to-end prototype	Front-end review behavior depends on consistent backend outputs.
Validation thresholds + held-out manual set	Final performance reporting	Thresholds must be frozen before unbiased final evaluation.

This is an architectural dependency view, **not** Member 2's Work Breakdown Structure, dependency schedule, detailed resource plan or risk register.

13. Conclusion

SmartListing Copilot has a well-bounded project concept: a single public fashion dataset, two visual attributes, parallel CV/NLP analysis, explainable mismatch review and human-controlled corrections. The revised planning analysis identifies a coherent problem and plausible technical approach without claiming verified performance or operational benefit. The principal decisions to validate next are the quality and separability of five categories, the reliability of type/color predictions, the realism of constructed listing text and the adequacy of integration and computing resources. Delivery should prioritize the core review pipeline and preserve the approved scope throughout the four university milestones.

14. References

[1] *SmartListing Copilot - Project Proposal (Revised Scope: Fashion Product Images Small).* Internal project proposal, 12 October 2026. Supplied authoritative project source.

[2] *Project Proposal: Analysis & Design Phase Blueprint.* Supplied academic guidance document, undated. Used for section-structure guidance only; unrelated example data excluded.

[3] P. Aggarwal, *Fashion Product Images (Small).* Kaggle. kaggle.com/paramaggarwal/fashion-product-images-small (accessed October 2026).

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," *Proc. IEEE CVPR*, pp. 770-778, 2016. doi:10.1109/CVPR.2016.90.

[5] A. G. Howard *et al.*, "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," arXiv:1704.04861, 2017.

[6] A. Radford *et al.*, "Learning Transferable Visual Models From Natural Language Supervision," *Proc. ICML*, 2021. arXiv:2103.00020.

[7] FastAPI, *Documentation.* https://fastapi.tiangolo.com/.

[8] Streamlit, *Documentation.* https://docs.streamlit.io/.

Reference use: [1] defines the project and exclusions; [2] informs analysis structure; [3] identifies the data source; [4-6] substantiate general model-family availability, not project-specific effectiveness; [7-8] identify the planned framework documentation.

SmartListing Copilot

Multimodal Listing Quality Control for Fashion E-Commerce

Project Management Document

Work Breakdown Structure | Gantt Chart | Timeline and Milestones | Team Roles | Resources | Weekly Plan | Risk Register and Mitigation

Deliverable	Member 2: Project Management
Version / status	1.0, planning baseline for team review; no supervisor approval is claimed
Prepared	15 October 2026, for submission on 16 October 2026
Planning horizon	19 October to 4 December 2026 (7 working weeks, 6 technical roles, 588 forward hours)
Scope authority	SmartListing Copilot Project Proposal (Revised Scope), consistent with the Project Planning Document and the Functional Requirements Specification

Official submission dates (DEPI schedule)

16 Oct 2026	6 Nov 2026	30 Nov 2026	4 Dec 2026
Planning, Literature Review, Requirements	System Analysis and Design	Implementation (source code and execution)	Final Presentation, Testing and Reports

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1. Document Overview

This document is the project-management baseline that lets a six-person team deliver SmartListing Copilot by the four official university dates: 16 October, 6 November, 30 November and 4 December 2026.

Purpose. It turns the approved scope into an executable plan: a Work Breakdown Structure (WBS), a drawn Gantt chart, a timeline and milestones locked to the official dates, six balanced technical roles, a resource plan, a week-by-week plan, and a full risk register with mitigation strategies.

Scope assurance. The plan covers only what the Revised Scope fixes [1][2][3]. Fashion Product Images Small is the sole MVP dataset. Five fashion product types are chosen after EDA. Only Product Type and Color are checked. Computer vision (CV) and NLP run in parallel. Every suggested change needs the seller's Accept / Edit / Ignore decision, and nothing is published automatically. No out-of-scope capability (price checks, LLM text generation, batch processing, live marketplace integration) is planned.

1.1 Document control

Field	Value
Project	SmartListing Copilot: Multimodal Listing Quality Control for Fashion E-Commerce
Document	Project Management Document (WBS, Gantt Chart, Timeline, Milestones, Roles, Risk Register)
Responsible	Member 2: Project Management
Version	1.0, planning baseline
Submission date	16 October 2026 (official planning milestone)
Planning horizon	19 October to 4 December 2026 (7 working weeks)
Status	Plan for team review; no supervisor approval is claimed

1.2 Evidence labels and source hierarchy

Three labels are used, matching the Member 1 planning document [2]. **Confirmed** means fixed by the official schedule or by the scope documents. **Proposed** means a plan that has not yet been carried out. **Assumption** means a condition that must be verified later. Effort figures in this document are planning values, not measured actuals, and no performance result is claimed.

When sources differ, they are applied in this order:

1. The official project documentation schedule [4] governs all dates.
2. The Revised Scope proposal [1] governs scope. It was not supplied separately, so its content is taken as quoted in [2] and [3].
3. The Member 1 planning document [2] governs objectives OBJ-01 to OBJ-09 and success criteria SC-01 to SC-10.

4. The Member 5 Functional Requirements Specification [3] governs FR-001 to FR-025, AC-001 to AC-025 and the open decisions.
5. The project-management brief [5] governs which items this document must contain.

1.3 How this document answers the brief

Required item	Section
Work Breakdown Structure (WBS)	3
Project Timeline	4
Milestones	5
Gantt Chart	6
Team Roles and Responsibilities	7
Required Resources and Technologies	8
Weekly Plan and Deliverables	9
Risk Management Plan	10
Risk Register	11
Risk Mitigation Strategies	12

Section 13 links the WBS to the objectives, functional requirements and success criteria. Section 14 records assumptions and the decisions that are still open. The four deliverables requested in the brief are all inside this document: the Work Breakdown Structure (section 3), the Gantt chart (section 6), the project timeline and milestones (sections 4 and 5) and the Risk Register with mitigation strategies (sections 11 and 12).

2. Project Management Approach

The project runs as four gated phases fixed by the official dates, with one-week iterations inside each phase and a stub-first integration strategy.

2.1 Delivery model

The four official dates cannot move, so the top level is plan-driven: Planning (16 Oct), Analysis and Design (6 Nov), Implementation (30 Nov), and Testing and Final Presentation (4 Dec). Inside each phase the team works in one-week iterations that end with a status checkpoint. Integration is stub-first: every module publishes a stub that already returns the agreed data structure, so the API and user interface can be built in parallel with the models and replaced by real modules one at a time. This follows the PMBOK Guide idea of tailoring the approach to the project [7] and the Scrum Guide idea of short time-boxed iterations that end in a checkpoint [8].

The plan protects the core pipeline above everything else. The core is image plus text input, CV and NLP attribute extraction, comparison with reason codes and severity, the three-state Review Queue, controlled suggestions, and seller Accept / Edit / Ignore. Supporting features are cut first if the schedule is threatened (section 12.4).

2.2 Governance and decision rights

Decision	Who decides	Rule	Recorded in
Technical choice inside one module	Role lead, after peer-buddy review	Choice must respect the shared data contract and the scope	Pull request and board card
Decision gates DG-1 to DG-3	Whole team	Lead presents evidence; consensus, otherwise majority of six with the PM and Integration Lead breaking a tie	Gate record in the change log
Scope change request	Whole team, supervisor consulted if the scope is affected	Written request with time, hours and risk impact; default answer is no	Change log
Activating a risk contingency or the cut order	PM and Integration Lead with the risk owner	Trigger in the risk register has been met	Risk register and change log
Resolving an open requirement decision (Q-01, Q-02, IN-01 to IN-04)	Team, then supervisor confirmation if available	Decision, evidence and date are written down before any rule is treated as accepted	Section 14 and the FRS change log [3]

2.3 Communication plan

Event	Frequency	Participants	Output
Weekly sync (30 to 45 minutes)	Once a week, start of week	All six roles	Plan for the week, risk review, decisions needed, minutes
Written mid-week status	Once a week	All six roles, one short message each	Green, Amber or Red per owned package
Peer-buddy review	Within one working day of a review request	The two buddies	Approved or returned pull request or document

Event	Frequency	Participants	Output
Gate review	At DG-1 to DG-3, IM-1 to IM-3 and each official milestone	All six roles	Gate record with pass, conditional pass or fail
Supervisor update, if applicable	Before each official milestone	PM and Integration Lead	Short status and list of open decisions

Channels: one team chat group, one shared task board, one shared Git repository for code, and one shared folder for documents. Every file name carries its WBS or document ID and a version number.

2.4 Tracking and reporting

Each WBS package is one board card with its WBS ID, lead role, peer buddy, planned hours and dates. Cards move through Backlog, In progress, In review and Done.

Status	Meaning
Green	On plan, no blocking dependency
Amber	Slipping by up to one working day, or waiting on a dependency
Red	Slipping by more than one working day on the critical path, or blocking a gate

The weekly one-page status report shows progress against the plan, hours used against planned hours per role, changes in the risk register, open decisions, and from week 6 the number of acceptance fixtures passing.

2.5 Definitions of Ready and Done

- **Ready:** predecessors are done, required inputs or data contracts exist, and the test cases or checks that will prove the package are identified.
- **Done:** the output is committed with its WBS ID; checks or unit tests pass; the peer buddy has reviewed it; the README or docstring is updated; the output works against the shared data contract; and the card is moved to Done.

2.6 Quality assurance through peer buddies

Every deliverable has an author and a reviewer from a fixed buddy pair, so each role both gives and receives exactly one review stream.

Buddy pair	Why this pairing
Deep Learning / Computer Vision and Machine Learning	Models and data share splits, labels, thresholds and calibration
NLP and Rules and Decision Logic	Text claims feed the comparison rules and use the same vocabulary
Frontend and QA and Project Manager and Integration	The interface and tests depend on the API contract and integration order

2.7 Change control and scope guard

Any addition, removal or change of scope is raised as a written change request that states the reason, the WBS packages affected, the hours and dates affected, and the risks affected. Requests are decided at the weekly sync and logged. The out-of-scope list in [2] and [3] is a hard boundary: return prediction, price validation, LLM generation, material or size recognition, batch processing, live marketplace integration and production cloud deployment are not added by change request.

2.8 Project rules that protect evaluation integrity

These rules come from the planning and requirements documents [2][3] and apply to every role.

1. Split by product ID into train, validation and test before any synthetic error is injected.
2. Freeze thresholds on validation data (gate DG-3) before the held-out evaluation starts; never tune on the test or manual sets.
3. Report results from synthetic error cases and from the manually constructed set separately.
4. Mark every evaluation description as manually written or synthetic, with its origin.
5. Treat a proposed rule as a proposal, not an accepted rule, until the decision, its evidence and its date are recorded.

3. Work Breakdown Structure (WBS)

The WBS divides the project into five level-1 areas and 44 forward work packages; every package has one lead role, dates, planned hours and a link to the objectives, requirements and success criteria it serves.

Role tags used in all tables. Roles are technical roles, not member numbers.

Tag	Role
DL/CV	Deep Learning and Computer Vision Engineer
ML	Machine Learning Engineer
NLP	NLP Engineer
RDL	Rules and Decision Logic Engineer
FQA	Frontend and QA Engineer
PMI	Project Manager and Integration Lead

3.1 Level-1 areas

ID	Area	Window	Official date	Packages	Planned hours
1.0	Planning baseline: Project Planning, Literature Review, Requirements, Project Management documents	12 to 16 Oct	16 Oct	4 documents (due 16 Oct)	Not counted; completed before 19 Oct
2.0	System Analysis and Design	19 Oct to 6 Nov	6 Nov	12	174
3.0	Implementation (source code and execution)	9 Nov to 30 Nov	30 Nov	18	232
4.0	Testing, evaluation and final delivery	23 Nov to 4 Dec	4 Dec	10	118
5.0	Project management and governance	19 Oct to 4 Dec	Continuous	4	64
	Total forward effort			**44**	**588**

3.2 Area 2.0: System Analysis and Design (19 Oct to 6 Nov)

ID	Work package	Lead	Dates	Hours	Output	Traces to
2.1	Data audit and EDA	ML	19 to 23 Oct	20	Audit of files, IDs, missing metadata, corrupt images, duplicates, class and color distributions, label quality; EDA report	OBJ-02; FR-004, FR-005

ID	Work package	Lead	Dates	Hours	Output	Traces to
2.2	Five-class selection and freeze (gate DG-1)	ML	26 to 28 Oct	12	Frozen list of exactly five articleType classes chosen for balance, visual separability, image availability and label quality	OBJ-02; FR-004, FR-015
2.3	Type and color vocabulary, groups and color families	RDL	22 to 30 Oct	14	Canonical type and color names with aliases, product-type groups, color families and near-shade table (v1)	OBJ-03, OBJ-04; FR-007, FR-010, FR-011, FR-017
2.4	Split protocol and leakage controls	ML	26 to 28 Oct	6	Product-ID split rule, duplicate checks across splits, order of error injection	OBJ-09; SC-09
2.5	CV design and baseline plan	DL/CV	22 to 30 Oct	17	Preprocessing, ResNet or MobileNet selection criteria, CLIP zero-shot prompts including the other-product prompt, image-quality and dominant-color check design	OBJ-02; FR-004, FR-005, FR-006
2.6	NLP design	NLP	20 to 30 Oct	15	Normalization steps, English rules and fuzzy matching, fixed Arabic lexicon, Arabic-script routing	OBJ-03; FR-007, FR-008
2.7	Comparison and decision-logic design	RDL	26 Oct to 4 Nov	15	Rule table for consistency, severity, reason codes, inconsistency level and queue routing; proposals for Q-01 and Q-02; design of optional score and missing-attribute checks	OBJ-04, OBJ-05, OBJ-06; FR-009 to FR-012, FR-014 to FR-020
2.8	Error-injection and mismatch-classifier design	ML	29 Oct to 4 Nov	14	Injection scenarios (wrong type, wrong color, combined, correct, near shade, missing claim), classifier features, comparison plan against the rule baseline	OBJ-04; FR-009
2.9	Architecture and API contract design	PMI	20 Oct to 2 Nov	18	Component architecture, Pydantic data contract v1.0 (frozen at DG-2), FastAPI endpoints, stub plan, repository layout	OBJ-01
2.10	UI wireframes and test-fixture design	FQA	26 Oct to 4 Nov	12	Streamlit wireframes for intake, results, queue, Before/After and Accept / Edit / Ignore; fixture list for AC-001 to AC-025; proposal for how seller decisions are represented	OBJ-08; FR-001 to FR-003, FR-022 to FR-025
2.11	Evaluation and test protocol	FQA	28 Oct to 4 Nov	14	Metrics, threshold-freeze procedure, manual-set specification and labeling guide	OBJ-09; SC-02, SC-03, SC-09, SC-10
2.12	Analysis and Design document and submission	PMI	2 to 6 Nov	17	Consolidated deliverable; internal freeze 5 Nov; submission 6 Nov	Milestone M1

3.3 Area 3.0: Implementation (9 Nov to 30 Nov)

ID	Work package	Lead	Dates	Hours	Output	Traces to
3.1	Data pipeline	ML	9 to 12 Nov	13	Cleaned metadata, product-ID train / validation / test splits, data loaders, normalized labels, passed leakage check	OBJ-02, OBJ-09
3.2	CLIP zero-shot baseline	DL/CV	9 to 13 Nov	7	Zero-shot type and color predictions with confidence; fallback path	OBJ-02; FR-004
3.3	Transfer-learning training for type and color	DL/CV	12 to 23 Nov	19	Trained ResNet or MobileNet models, saved weights, validation metrics	OBJ-02; FR-004, FR-005, FR-013
3.4	Image-quality, dominant-color and scope signals	DL/CV	16 to 20 Nov	12	Resolution, blur and brightness checks; dominant-color cross-check; CV versus CLIP disagreement signal	OBJ-02, OBJ-06; FR-005, FR-006, FR-015
3.5	Calibration and threshold selection (gate DG-3)	ML	20 to 25 Nov	14	Confidence calibration (ECE) and validation-derived thresholds frozen at DG-3	OBJ-02, OBJ-09; FR-013, FR-014
3.6	NLP claim extraction, Arabic lexicon and script routing	NLP	9 to 18 Nov	16	Structured type and color claims from title and description	OBJ-03; FR-007, FR-008

ID	Work package	Lead	Dates	Hours	Output	Traces to
3.7	Synthetic error injection and mismatch dataset	ML	12 to 18 Nov	10	Labeled mismatch cases generated after splitting, with provenance flags	OBJ-04, OBJ-09; SC-03, SC-09
3.8	Lightweight mismatch classifier	ML	18 to 24 Nov	11	Logistic regression or small MLP compared with the rule baseline	OBJ-04; FR-009
3.9	Comparison engine, severity and reason codes	RDL	9 to 18 Nov	18	Type, color and missing-claim comparison, severity table, reason codes, explanation templates	OBJ-04, OBJ-05; FR-009 to FR-012, FR-016, FR-017
3.10	Inconsistency level, queue routing and supporting features	RDL	16 to 23 Nov	14	Listing Inconsistency Level, three-state Review Queue, low-confidence and unsupported-category handling, optional Listing Quality Score	OBJ-06; FR-014, FR-015, FR-018 to FR-020
3.11	Template-based suggested fix	NLP	16 to 24 Nov	13	Type-and-color-only title and description templates with Before/After difference data	OBJ-07; FR-021, FR-022
3.12	Manual evaluation set	NLP	13 to 25 Nov	15	Independent set of English and Arabic listings built from test-split images (proposed target 120, within the planned 100 to 150), checked by a second person	OBJ-03, OBJ-09; SC-09
3.13	FastAPI backend and data contract implementation	PMI	9 to 18 Nov	16	Orchestration of CV and NLP branches, Pydantic schemas, stub endpoints live by 13 Nov, API usage notes	OBJ-01
3.14	Streamlit interface: intake and results	FQA	10 to 18 Nov	13	Intake form, findings view with reason codes and confidence, Review Queue view	OBJ-01, OBJ-05, OBJ-06; FR-001 to FR-003
3.15	Streamlit interface: Before/After and seller decision	FQA	18 to 25 Nov	11	Side-by-side view with reasons for changes; Accept, Edit and Ignore with no publish action	OBJ-07, OBJ-08; FR-022 to FR-025
3.16	Integration and end-to-end smoke tests	PMI	13 to 27 Nov	19	Stubs replaced by real modules; gates IM-1 (13 Nov), IM-2 (20 Nov), IM-3 (27 Nov)	OBJ-01; SC-01
3.17	Reproducibility package and optional MLflow and Docker	PMI	23 to 27 Nov	9	Requirements file, README, notebooks, saved weights, API notes; MLflow and Docker only if the core is secure	OBJ-09
3.18	Code freeze and Implementation submission	PMI	27 to 30 Nov	2	Frozen repository tag; source code and run instructions submitted 30 Nov	Milestone M2

3.4 Area 4.0: Testing, evaluation and final delivery (23 Nov to 4 Dec)

ID	Work package	Lead	Dates	Hours	Output	Traces to
4.1	Acceptance test execution (AC-001 to AC-025)	FQA	23 Nov to 1 Dec	16	Test log with pass or fail per AC, defect list	OBJ-01, OBJ-08; SC-01, SC-04, SC-07
4.2	Held-out synthetic evaluation	ML	26 Nov to 1 Dec	16	Accuracy, macro-F1, confusion matrix and ECE for type and color; precision, recall, F1 and false-alarm rate for mismatches	OBJ-02, OBJ-04, OBJ-09; SC-02, SC-03
4.3	Manual-set evaluation	NLP	26 Nov to 1 Dec	15	Claim precision, recall and F1 by claim and language; mismatch metrics reported separately from synthetic results	OBJ-03, OBJ-09; SC-09
4.4	Routing and uncertain-case scenario tests	RDL	24 to 30 Nov	14	Scenario checks of the three queue states, LOW_CONFIDENCE, IMAGE_BLUR and UNSUPPORTED_CATEGORY	OBJ-06; SC-05, SC-08

ID	Work package	Lead	Dates	Hours	Output	Traces to
4.5	End-to-end catch rate and response time	PMI	30 Nov to 1 Dec	12	Measured catch rate and response time	OBJ-09; SC-10
4.6	Confidence and error analysis	DL/CV	1 to 2 Dec	13	Confidence behavior, error cases and limitations	OBJ-09; SC-02, SC-10
4.7	Defect fixing and regression	Module owners, QA triages	27 Nov to 2 Dec	9	Bug-fix-only window and regression run; no new features	SC-01 to SC-10
4.8	Final testing and evaluation report	PMI	25 Nov to 3 Dec	10	Report on evaluation, threshold selection, data provenance and limitations	OBJ-09; SC-10
4.9	Final presentation and demo	PMI	30 Nov to 3 Dec	9	Slides, demo script, two rehearsals, backup recording	Milestone M3
4.10	Final submission package check	PMI	3 to 4 Dec	4	Checklist of every deliverable; submission 4 Dec	Milestone M3

3.5 Area 5.0: Project management and governance (19 Oct to 4 Dec)

ID	Work package	Lead	Dates	Hours	Output	Traces to
5.1	Weekly sync, status and reporting	PMI	19 Oct to 4 Dec	28	Minutes and one-page weekly status	All objectives
5.2	Risk register upkeep and weekly risk review	PMI	19 Oct to 4 Dec	16	Updated register, escalations	Section 11
5.3	Peer review and quality gates	PMI	19 Oct to 4 Dec	14	Review records and gate records	Section 5
5.4	Change control, scope guard, document and version control	PMI	19 Oct to 4 Dec	6	Change log, repository and board setup in week 1, document versions	Section 2

Area 1.0 contains the four documents submitted on 16 October: 1.1 Project Planning, 1.2 Literature Review, 1.3 Requirements Gathering (Functional Requirements Specification) and 1.4 this Project Management Document.

4. Project Timeline

The plan runs for seven working weeks, from 19 October to 4 December 2026, and every phase finishes on or before its official date, with an internal freeze one working day earlier.

4.1 Alignment with the official schedule

The dates below follow the DEPI column of the official Project Documentation Guidelines [4]; the earlier DEPI Industry dates do not apply to this team.

Phase	Planning window	Official date	Internal freeze	Official content [4]	Working weeks
Planning baseline	12 to 16 Oct	Fri 16 Oct 2026	Final check Thu 15 Oct	Project Planning and Management, Literature Review, Requirements Gathering	Baseline
Analysis and Design	19 Oct to 6 Nov	Fri 6 Nov 2026	Thu 5 Nov	System Analysis and Design	Weeks 1 to 3
Implementation	9 to 30 Nov	Mon 30 Nov 2026	Fri 27 Nov	Implementation (source code and execution)	Weeks 4 to 6, plus Monday 30 Nov
Testing and Final Delivery	23 Nov to 4 Dec	Fri 4 Dec 2026	Thu 3 Dec	Final Presentation, Testing and Reports	Weeks 6 and 7

4.2 Calendar of working weeks

Week	Dates	Main phase
Week 1	19 to 23 Oct	Analysis and Design
Week 2	26 to 30 Oct	Analysis and Design
Week 3	2 to 6 Nov	Analysis and Design; submission 6 Nov
Week 4	9 to 13 Nov	Implementation
Week 5	16 to 20 Nov	Implementation
Week 6	23 to 27 Nov	Implementation completion; testing begins
Week 7	30 Nov to 4 Dec	Implementation submission 30 Nov; testing, evaluation and final delivery

4.3 Timeline rules

1. **Official dates are fixed.** No package is scheduled to finish after the official date of the phase it belongs to.

2. **One working day of buffer.** Each official date has an internal freeze one working day earlier, so that a late fix, an upload problem or a formatting error does not cause a missed submission.
3. **Testing overlaps implementation on purpose.** The official dates leave only four days between code submission (30 Nov) and the final presentation (4 Dec). Acceptance testing therefore starts on 23 Nov against modules as they become integrated, and the held-out evaluation starts on 26 Nov, one day after thresholds are frozen.
4. **Order that cannot change.** Classes are frozen (DG-1, 28 Oct) before splits are built; splits come before error injection; thresholds are frozen on validation data (DG-3, 25 Nov) before any held-out result is produced.
5. **Stub-first integration.** A working end-to-end skeleton with stubs is due on 13 Nov (IM-1), a first run with real modules on 20 Nov (IM-2), and feature completeness with code freeze on 27 Nov (IM-3).
6. **Bug-fix-only window.** From 27 Nov to 2 Dec only defects are fixed; no new feature is accepted.

4.4 Effort and time are matched

Each role is planned at 98 hours over the seven weeks, an average of 14 hours per week. This is a planning assumption to be confirmed with the team in week 1 (section 14); section 7.3 shows how the hours are balanced across the six roles. The tight gap between 30 Nov and 4 Dec is the main schedule exposure and is covered by risk R-14.

5. Milestones

Four official milestones and six internal gates control the project; each has a date, exit criteria and a named owner.

5.1 Official milestones

The entry criterion of each milestone is the exit of the previous one. The Project Manager and Integration Lead owns every milestone check.

ID	Milestone	Official date	Internal freeze	Exit criteria	WBS
M0	Planning baseline submitted	Fri 16 Oct	Thu 15 Oct	Project Planning, Literature Review, Requirements (FRS) and Project Management documents submitted; scope, OBJ-01 to OBJ-09 and FR-001 to FR-025 baselined	1.0
M1	Analysis and Design submitted	Fri 6 Nov	Thu 5 Nov	EDA report and frozen five classes; dataset schema and split protocol; system design with data contract v1.0; model baseline plan; evaluation protocol; open decisions Q-01, Q-02 and IN-01 to IN-04 recorded	2.1 to 2.12
M2	Implementation submitted	Mon 30 Nov	Fri 27 Nov	Cleaned data pipeline; CV model and CLIP baseline; NLP claims; mismatch rules and classifier; Review Queue and templates; integrated FastAPI and Streamlit prototype that runs from the submitted code with run instructions	3.1 to 3.18
M3	Testing, final presentation and reports	Fri 4 Dec	Thu 3 Dec	Held-out synthetic and manual evaluation results; confidence and error analysis; acceptance test log; reproducibility materials; demo; final documentation	4.1 to 4.10

5.2 Internal decision and integration gates

ID	Gate	Date	What is decided or checked	Owner	If missed
DG-1	Five classes frozen	Wed 28 Oct	Exactly five articleType classes accepted on class balance, visual separability, image availability and label quality	ML presents, team decides	Freeze on Fri 30 Oct with the best available evidence; no later change without a change request
DG-2	Interfaces frozen	Mon 2 Nov	Data contract v1.0, type and color vocabulary and color-family tables frozen	PMI and RDL	Freeze with documented open fields; stubs return defaults for them
IM-1	Stub-based end-to-end skeleton	Fri 13 Nov	Interface, API, stubs, Review Queue and seller decision work together on fixed sample data	PMI	Daily integration pair sessions until met; interface work continues on stubs
IM-2	First real end-to-end run and scope-cut checkpoint	Fri 20 Nov	CV, NLP, comparison and queue run on real modules for at least one listing; decide whether to cut supporting features	PMI	Apply cut number 1 immediately (section 12.4)
DG-3	Thresholds frozen	Wed 25 Nov	Confidence, blur, near-shade and classifier thresholds fixed on validation data; manual set locked read-only	ML presents, FQA audits	Use last validation-derived values and report the calibration as incomplete
IM-3	Feature complete and code freeze	Fri 27 Nov	All core packages integrated; reproducibility package present; no Red item on the core	PMI	Apply the next cut in the order; the bug-fix-only window starts regardless

The open requirement decisions have their own deadline, Wed 4 Nov, so that they are recorded before the 5 Nov freeze (section 14).

5.3 Gate result

Every gate ends in one of three results, written in the change log: **pass**; **conditional pass**, with the condition and a due date no later than two working days away; or **fail**, which triggers the cut order or a change request.

6. Gantt Chart

The Gantt chart is drawn at two levels from the dates and hours in section 3 and the gates in section 5. Figure 1 is the master chart: it shows the whole project from 12 October to 4 December on one page, with the four official milestones, the phases, the internal gates and the critical path. Figures 2 to 4 are detailed charts with one bar for each of the 44 forward work packages, one figure for each phase. Every symbol, colour and line used in the charts is explained in section 6.1, and each figure is explained row by row after it.

6.1 How to read the charts

Element	Appearance	Meaning
Day columns	Thin vertical columns; weekday letter and date number at the top	Each column is one calendar day. Saturdays and Sundays are shaded grey. No work is planned on them, so they act as slack for late fixes.
Week brackets	"Week 1" to "Week 7" (W1 to W7 in Figure 1) above the dates	The seven working weeks, Monday to Friday, from 19 October to 4 December. The week numbers are the same as in the weekly plan (section 9).
Task bar	Rounded horizontal bar, one for each WBS package	Starts on the first working day of the package and ends at the end of its last working day. Bar length shows duration, not effort. Effort is given as hours in the Hours column.
Bar colour	Blue, orange, teal, navy, pink, green, grey	Lead role of the package: DL/CV blue, ML orange, NLP teal, RDL navy, FQA pink, PMI green. Grey is used for package 4.7 (defect fixing), which belongs to whichever role owns the module with the defect.
Lead and Hours columns	Text to the right of each detailed chart	The role that leads the package and its planned hours. The hours add up to 174, 232 and 118 for areas 2.0, 3.0 and 4.0 (section 3.1).
Official date line	Solid red vertical line	The end of an official submission day: 6 Nov (M1), 30 Nov (M2) and 4 Dec (M3). Nothing is planned to run past this line for its phase.
Gate line	Dashed vertical line with a boxed label	An internal gate from section 5.2: decision gates DG-1 (28 Oct), DG-2 (2 Nov) and DG-3 (25 Nov); integration gates IM-1 (13 Nov), IM-2 (20 Nov) and IM-3 (27 Nov).
Freeze line	Dotted grey vertical line	An internal freeze, one working day before an official date: 5 Nov, 27 Nov and 3 Dec. Content is frozen and used for the final checks and upload.
Diamonds (Figure 1)	Black, white, yellow and blue diamonds	Black: official milestone M0 to M3. White: internal freeze. Yellow: decision gate DG. Blue: integration gate IM.
Critical-path bars (Figure 1)	Red bars	The chain of packages where a delay moves a deliverable past its date (section 13.1).
Overlap bars (Figure 1)	Amber and dark grey bars	Amber: testing overlaps the last days of implementation (23 to 30 Nov). Dark grey: the bug-fix-only window (27 Nov to 2 Dec).

6.2 Master Gantt chart (Figure 1)

Figure 1. Master Gantt chart: official dates, phases, gates and critical path

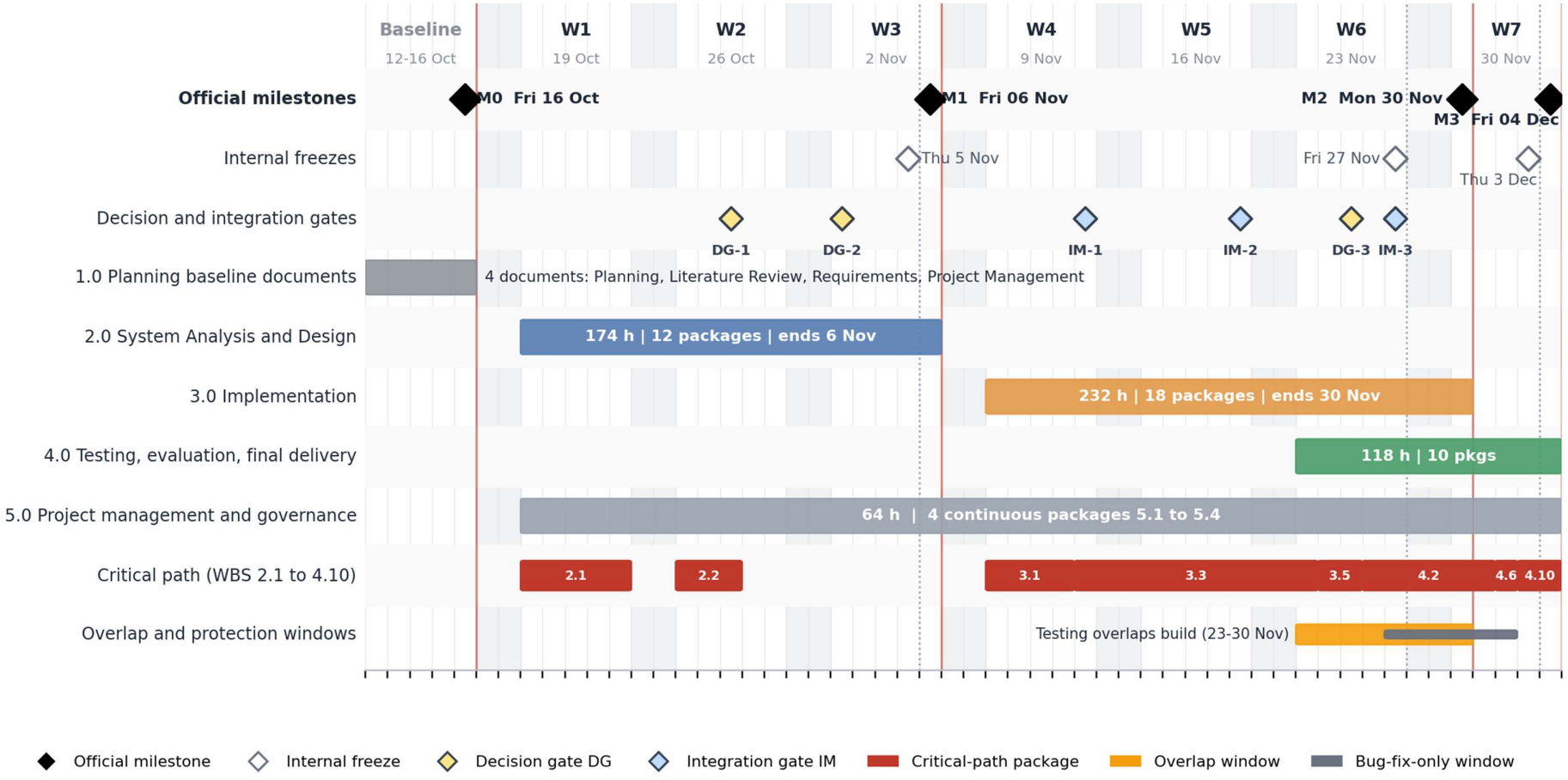


Figure 1 reads from top to bottom: first the fixed dates, then the control points, then the four phases, then the critical path, then the protection windows. The table explains each row.

Row	What is drawn	Dates	What it means for the team
Official milestones	Four black diamonds	M0 Fri 16 Oct; M1 Fri 6 Nov; M2 Mon 30 Nov; M3 Fri 4 Dec	The four official dates from the DEPI column of the guidelines [4]. All planning, design, code and test work is scheduled to finish on or before them.
Internal freezes	Three white diamonds	Thu 5 Nov; Fri 27 Nov; Thu 3 Dec	One working day before M1, M2 and M3 (Fri 27 Nov is the last working day before Mon 30 Nov). After a freeze only checks, fixes to the submission itself and upload are allowed.
Decision and integration gates	Yellow (DG) and blue (IM) diamonds	DG-1 Wed 28 Oct; DG-2 Mon 2 Nov; IM-1 Fri 13 Nov; IM-2 Fri 20 Nov; DG-3 Wed 25 Nov; IM-3 Fri 27 Nov	DG-1 freezes the five classes, DG-2 freezes the data contract and vocabulary, IM-1 proves the stub-based skeleton, IM-2 is the first real end-to-end run and the scope-cut checkpoint, DG-3 freezes thresholds on validation data, IM-3 is feature complete and code freeze.
1.0 Planning baseline	Grey bar	12 to 16 Oct	The four documents due on 16 October: Project Planning, Literature Review, Requirements (FRS) and this Project Management Document. Their effort is not counted in the 588 forward hours.
2.0 System Analysis and Design	Blue bar	19 Oct to 6 Nov	Twelve packages, 174 hours. Ends on M1. Contains DG-1 and DG-2 and the internal freeze on 5 Nov.
3.0 Implementation	Orange bar	9 to 30 Nov	Eighteen packages, 232 hours. Ends on M2. Contains IM-1, IM-2, DG-3 and IM-3 and the freeze on 27 Nov.
4.0 Testing, evaluation and final delivery	Green bar	23 Nov to 4 Dec	Ten packages, 118 hours. Ends on M3. It starts on 23 Nov, before implementation ends, because only four days separate code submission and the final presentation.
5.0 Project management and governance	Grey bar over the whole project	19 Oct to 4 Dec	Four continuous packages (5.1 to 5.4), 64 hours: weekly sync and status, risk register upkeep, peer review and gates, change control and version control.
Critical path	Red bars in sequence	2.1, 2.2, 3.1, 3.3, 3.5, 4.2, 4.6, 4.10	The sequence with almost no slack. Two bars are drawn from the day after the previous bar ends (3.3 from 13 Nov, 3.5 from 24 Nov) so that the chain does not overlap itself; their real windows are in section 3. Packages 2.4 and 4.8 are on the path in section 13.1 but run beside 2.2 and 4.2 to 4.6, so they are not drawn again.
Overlap and protection windows	Amber bar and dark grey bar	23 to 30 Nov; 27 Nov to 2 Dec	Amber: acceptance testing starts against integrated modules while the last build packages finish. Dark grey: from the code freeze on 27 Nov only defects are fixed and no new feature is accepted.

6.3 Detailed Gantt chart: Area 2.0 System Analysis and Design (Figure 2)

Figure 2. Detailed Gantt chart: Area 2.0 System Analysis and Design

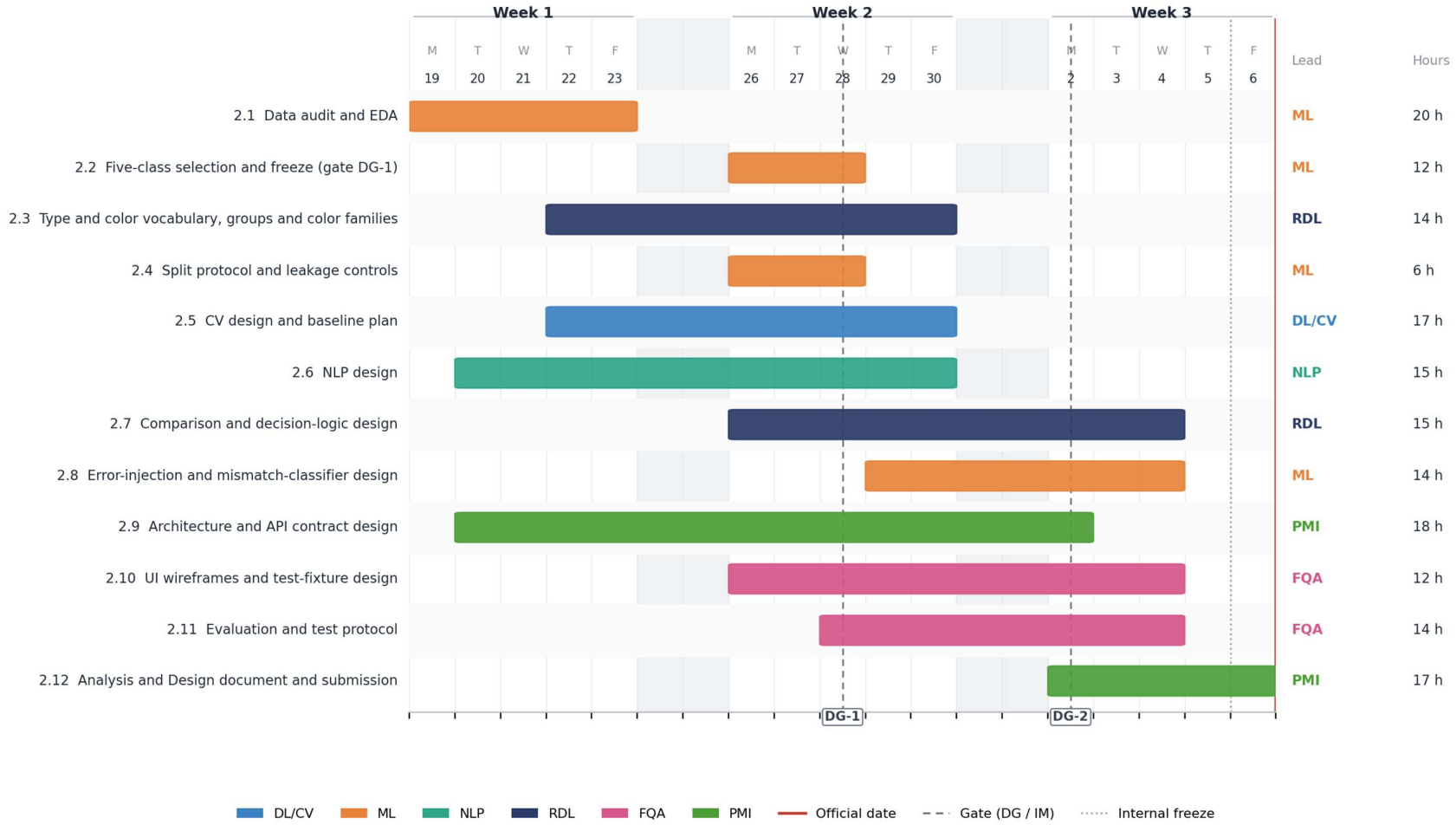


Figure 2 shows the three weeks before M1. The first week is data audit and setup, the second week freezes the classes and the design inputs, and the third week completes the design and submits it. Gate DG-1 falls on the last day of packages 2.2 and 2.4, and DG-2 falls on the last day of package 2.9.

Group	Packages	Window	Hours	What the bars show
Data and class selection	2.1, 2.2, 2.4	19 Oct to 28 Oct	38	EDA (2.1) comes first; five-class selection (2.2) and the split protocol (2.4) start only after the EDA draft on 23 Oct and end together with DG-1 on Wed 28 Oct. They are the first links of the critical path.
Vocabulary and model design inputs	2.3, 2.5, 2.6	20 Oct to 30 Oct	46	Vocabulary and color families (2.3), CV design (2.5) and NLP design (2.6) run in parallel and finish on Fri 30 Oct, so that version 1 of the shared tables is ready before DG-2.
Decision logic and error injection	2.7, 2.8	26 Oct to 4 Nov	29	Comparison and decision-logic design (2.7) and error-injection and classifier design (2.8) both end on Wed 4 Nov, the day by which the open decisions Q-01, Q-02 and IN-01 to IN-04 must be recorded.
Interfaces and test design	2.9, 2.10, 2.11	20 Oct to 4 Nov	44	The architecture and data contract (2.9) ends at DG-2 on Mon 2 Nov. UI wireframes and fixtures (2.10) and the evaluation protocol (2.11) end on Wed 4 Nov.
Consolidation and submission	2.12	2 Nov to 6 Nov	17	The Analysis and Design document is assembled from Mon 2 Nov, frozen on Thu 5 Nov and submitted on Fri 6 Nov (M1).

6.4 Detailed Gantt chart: Area 3.0 Implementation (Figure 3)

Figure 3. Detailed Gantt chart: Area 3.0 Implementation



Figure 3 is the busiest chart because all six roles build in parallel. Three integration gates cut through it: IM-1 on 13 Nov (stub-based skeleton), IM-2 on 20 Nov (first real run and scope-cut checkpoint) and IM-3 on 27 Nov (feature complete). DG-3 on 25 Nov is the date after which no threshold may change.

Group	Packages	Window	Hours	What the bars show
Data and CV model chain	3.1, 3.2, 3.3, 3.4, 3.5	9 Nov to 25 Nov	65	Data pipeline and splits (3.1) feed CLIP baseline (3.2) and training (3.3). Training is the longest package on the critical path (8 working days) and its main driver. Calibration and thresholds (3.5) need the final model outputs and end with DG-3 on Wed 25 Nov.
Text branch	3.6, 3.11, 3.12	9 Nov to 25 Nov	44	NLP claim extraction (3.6) comes first. Suggestion templates (3.11) and the manual evaluation set (3.12) follow; the manual set is complete on 25 Nov so it can be locked at DG-3.
Mismatch data and classifier	3.7, 3.8	12 Nov to 24 Nov	21	Error injection (3.7) starts only after the splits exist (3.1). The classifier (3.8) is trained on that data and compared with the rule baseline.
Decision layer	3.9, 3.10	9 Nov to 23 Nov	32	The comparison engine, severity and reason codes (3.9) come first; queue routing and the supporting features (3.10) use their output.
Application layer	3.13, 3.14, 3.15	9 Nov to 25 Nov	40	The FastAPI backend (3.13, stub endpoints live by 13 Nov) and the intake and results screens (3.14) are built against the data contract. Before/After and seller decision (3.15) follow the templates.
Integration and release	3.16, 3.17, 3.18	13 Nov to 30 Nov	30	Integration (3.16) runs across gates IM-1, IM-2 and IM-3. The reproducibility package (3.17) is prepared in the last week. The code is frozen on Fri 27 Nov and submitted on Mon 30 Nov (M2).

6.5 Detailed Gantt chart: Area 4.0 Testing, evaluation and final delivery (Figure 4)

Figure 4. Detailed Gantt chart: Area 4.0 Testing, evaluation and final delivery

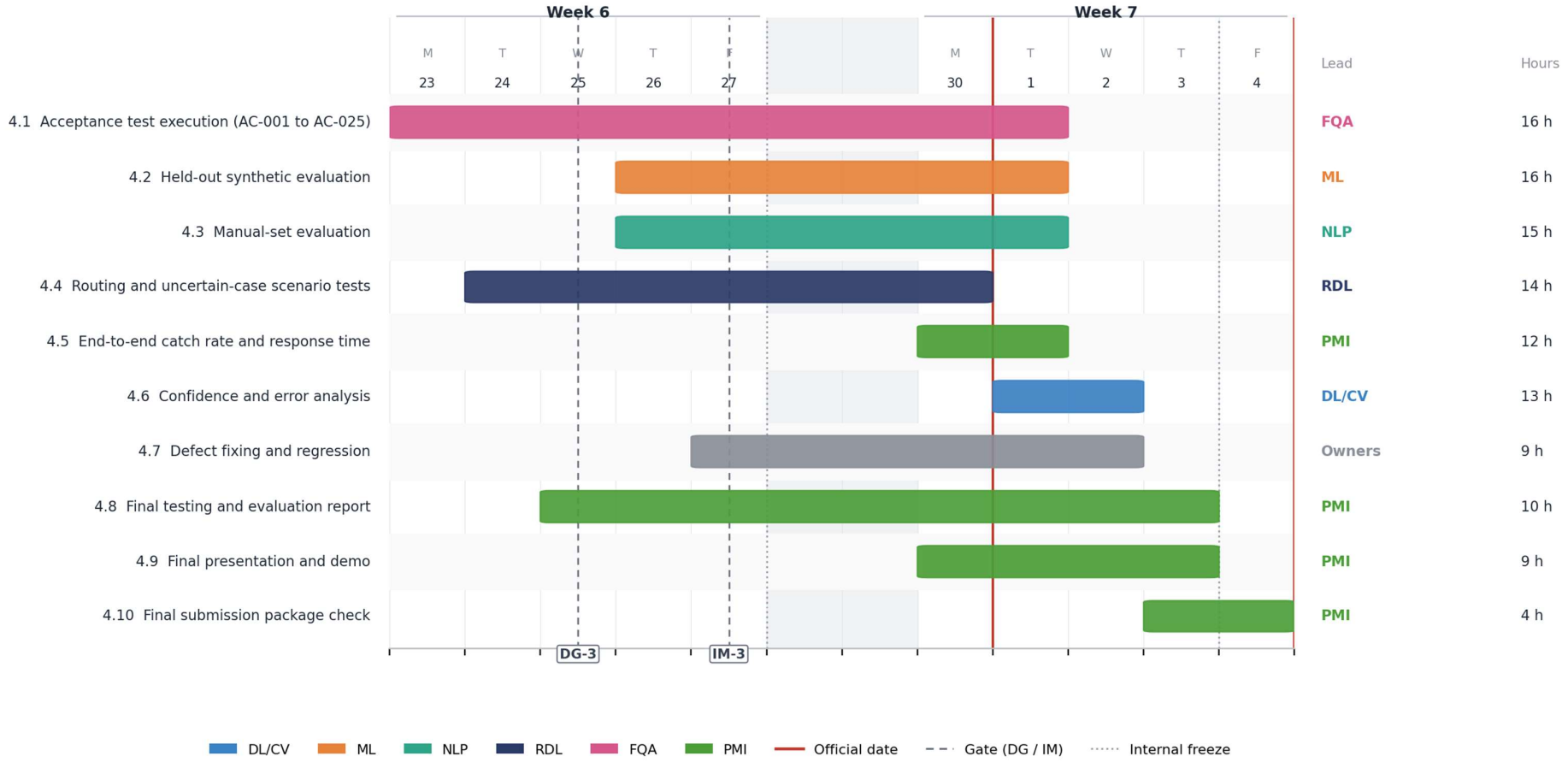


Figure 4 covers the last two weeks. Acceptance testing begins on 23 Nov against integrated modules. The held-out evaluation (4.2 and 4.3) starts on Thu 26 Nov, the day after DG-3, so that no held-out result exists before thresholds are frozen. The official date is Fri 4 Dec and the internal freeze is Thu 3 Dec.

Group	Packages	Window	Hours	What the bars show
Acceptance and routing tests	4.1, 4.4	23 Nov to 1 Dec	30	Execution of AC-001 to AC-025 (4.1) and scenario tests of the three queue states and the uncertain cases (4.4).
Held-out evaluation	4.2, 4.3	26 Nov to 1 Dec	31	Synthetic evaluation (4.2) and manual-set evaluation (4.3) both start the day after DG-3 and are reported separately.
Measurement and analysis	4.5, 4.6	30 Nov to 2 Dec	25	End-to-end catch rate and response time (4.5), then confidence and error analysis (4.6), which feeds the limitations in the report.
Defect window	4.7	27 Nov to 2 Dec	9	Bug-fix-only window from the code freeze on 27 Nov to 2 Dec. Whoever owns the module with the defect fixes it; FQA triages.
Reporting and presentation	4.8, 4.9, 4.10	25 Nov to 4 Dec	23	Final testing and evaluation report (4.8), presentation and demo with two rehearsals (4.9), and the submission package check (4.10) on 3 to 4 Dec.

Area 5.0 (package 5.1 to 5.4) is drawn only in Figure 1 because it runs for the whole project and has no single start or end inside a phase.

6.6 Critical path and slack

The critical path is the chain of packages whose delay would move a submission past its official date: data audit (2.1), class freeze (2.2) and split protocol (2.4), data pipeline (3.1), transfer-learning training (3.3), calibration and thresholds (3.5), held-out evaluation (4.2), confidence and error analysis (4.6), the final report (4.8) and the submission package check (4.10). It is listed step by step with the reason each step cannot slip in section 13.1.

Point on the path	Date	Slack and protection
Class freeze, DG-1	Wed 28 Oct	If the evidence is not ready, the freeze is made on Fri 30 Oct with the best available evidence (section 5.2). Later changes need a change request.
Interface freeze, DG-2 and M1	Mon 2 Nov; Fri 6 Nov	One working day of slack between the freeze on Thu 5 Nov and the submission on Fri 6 Nov.
Training and thresholds, DG-3	Wed 25 Nov	Training (3.3) ends on Mon 23 Nov and calibration (3.5) on Wed 25 Nov. Held-out evaluation starts on Thu 26 Nov, so the path has no spare day here; risks R-04 and R-13 protect it.
Code submission, M2	Mon 30 Nov	The code is frozen on Fri 27 Nov. Only defects are fixed afterwards.
Evaluation and report	1 to 3 Dec	Evaluations finish on 1 Dec, error analysis on 2 Dec and the report on 3 Dec.
Final submission, M3	Fri 4 Dec	One working day of slack: internal freeze on Thu 3 Dec, submission on Fri 4 Dec.

The overall slack of the plan is therefore about one working day before each official date. This is why the plan fixes the order of work that cannot change (section 4.3) and applies the cut order of section 12.4 as soon as a trigger is met.

7. Team Roles and Responsibilities

Six technical roles share the work equally: each role carries 98 planned hours, owns a distinct part of the pipeline, and has a fixed peer buddy.

The roles follow the technical structure of the system (image branch, data and evaluation, text branch, decision layer, interface and testing, integration), not member numbers. Any team member can be assigned to any role; the assignment is made once, in week 1.

7.1 Role definitions

Role	Core responsibility	Requirements (primary; shared)	Objectives	Peer buddy
Deep Learning and Computer Vision Engineer (DL/CV)	Builds and evaluates the image branch: preprocessing, ResNet or MobileNet transfer learning for Product Type and Color, CLIP zero-shot baseline and fallback, image-quality checks, dominant-color cross-check, CV confidence output	FR-004, FR-005, FR-006; shared FR-013, FR-015	OBJ-02	ML
Machine Learning Engineer (ML)	Owns data and evaluation science: EDA, five-class selection, label audit, product-ID splits, synthetic error injection, mismatch classifier, calibration and thresholds, held-out evaluation metrics	FR-013 (calibration), FR-014 (threshold); shared FR-009 (classifier)	OBJ-02, OBJ-04, OBJ-09	DL/CV
NLP Engineer (NLP)	Owns the text branch: normalization, English rules and fuzzy matching, fixed Arabic lexicon, script routing, controlled suggestion templates, manual evaluation set	FR-007, FR-008, FR-021, FR-022 (difference data); shared FR-016 (explanation wording)	OBJ-03, OBJ-07	RDL
Rules and Decision Logic Engineer (RDL)	Owns the decision layer: type and color vocabulary, groups and color families, comparison rules, severity, reason codes, inconsistency level, queue routing, missing-attribute check, optional Listing Quality Score	FR-009 to FR-012, FR-015 to FR-020; shared FR-014	OBJ-04, OBJ-05, OBJ-06	NLP
Frontend and QA Engineer (FQA)	Owns the Streamlit interface and quality: intake, results, queue, Before/After, Accept / Edit / Ignore, test fixtures, acceptance tests, evaluation protocol, usability walkthrough	FR-001 to FR-003 (interface), FR-023 to FR-025; shared FR-022	OBJ-01, OBJ-08	PMI
Project Manager and Integration Lead (PMI)	Owns planning, governance and integration: FastAPI backend, Pydantic data contract, orchestration, risk register, change control, reproducibility package, final report and presentation	API side of FR-001 to FR-003; end-to-end integration of all requirements	OBJ-01, OBJ-09 (reporting)	FQA

7.2 Ownership of packages and risks

Role	WBS packages led	Risks owned (section 11)
DL/CV	2.5, 3.2, 3.3, 3.4, 4.6	R-04, R-07, R-08, R-13, R-20, R-27
ML	2.1, 2.2, 2.4, 2.8, 3.1, 3.5, 3.7, 3.8, 4.2	R-01, R-02, R-05, R-11, R-28
NLP	2.6, 3.6, 3.11, 3.12, 4.3	R-09, R-10, R-24, R-29, R-30

Role	WBS packages led	Risks owned (section 11)
RDL	2.3, 2.7, 3.9, 3.10, 4.4	R-03, R-16, R-18, R-19, R-31
FQA	2.10, 2.11, 3.14, 3.15, 4.1	R-12, R-21, R-23, R-25, R-32
PMI	2.9, 2.12, 3.13, 3.16, 3.17, 3.18, 4.5, 4.8, 4.9, 4.10, 5.1 to 5.4	R-06, R-14, R-15, R-17, R-22, R-26

Package 4.7 (defect fixing) is led by whoever owns the module with the defect; FQA triages. The number of packages or requirements per role differs because they differ in size, so fairness is measured in hours (next table), not in counts.

7.3 Workload balance

Every role has the same 98 hours. The Project Manager and Integration Lead carries 16 more hours of governance work (area 5.0), so the technical load of that role is 16 hours lower than the others.

Role	2.0 Design	3.0 Implementation	4.0 Testing and delivery	5.0 Governance	Total hours
DL/CV	30	40	20	8	98
ML	30	40	20	8	98
NLP	30	40	20	8	98
RDL	30	40	20	8	98
FQA	30	40	20	8	98
PMI	24	32	18	24	98
Team	**174**	**232**	**118**	**64**	**588**

Figure 5. Planned hours per role and week (package hours spread evenly over each package's working days)

	Week 1 19 Oct	Week 2 26 Oct	Week 3 2 Nov	Week 4 9 Nov	Week 5 16 Nov	Week 6 23 Nov	Week 7 30 Nov	Total
DL/CV	10	18	5	14	22	12	17	98 h
ML	9	15	9	15	17	19	13	98 h
NLP	11	16	7	14	21	16	14	98 h
RDL	7	18	8	13	23	18	11	98 h
FQA	6	16	11	13	20	22	9	98 h
PMI	8	13	13	12	15	19	19	98 h
Team	52	96	53	80	117	106	83	

Figure 5 shows how the 98 hours of each role fall across the seven weeks. Each package's hours are spread evenly over its working days. The profile is uneven: weeks 1 and 3 are light (about 5 to 13 hours per role), and weeks 5 and 6 are heavy (up to 22 to 23 hours for some roles) because building and testing overlap. The average stays at 14 hours per week. Three consequences follow: availability for weeks 5 and 6 is confirmed in week 1 (assumption A-01); the light weeks are used for setup, peer review and tool practice inside the role's own packages; and if a role cannot give the hours in week 5 or 6, the Project Manager and Integration Lead rebalances them by change request or applies the cut order (risk R-17).

7.4 Allocation of hours by work package

Columns are roles; the largest figure in a row belongs to the lead role of that package.

ID	DL/CV	ML	NLP	RDL	FQA	PMI	Total
2.1	4	7	3	2	4	0	20
2.2	3	3	1	2	1	2	12

ID	DL/CV	ML	NLP	RDL	FQA	PMI	Total
2.3	1	1	5	7	0	0	14
2.4	1	3	0	0	2	0	6
2.5	13	2	0	1	0	1	17
2.6	0	0	10	2	2	1	15
2.7	1	1	2	8	2	1	15
2.8	1	7	3	2	1	0	14
2.9	2	1	2	2	1	10	18
2.10	0	0	1	1	9	1	12
2.11	2	3	1	1	6	1	14
2.12	2	2	2	2	2	7	17
2.0	**30**	**30**	**30**	**30**	**30**	**24**	**174**
3.1	2	7	1	1	2	0	13
3.2	6	1	0	0	0	0	7
3.3	15	3	0	0	1	0	19
3.4	8	1	0	2	1	0	12
3.5	3	6	0	3	1	1	14
3.6	0	1	11	2	2	0	16
3.7	0	5	3	1	1	0	10
3.8	1	7	0	2	1	0	11
3.9	1	1	2	11	2	1	18
3.10	0	1	1	9	2	1	14
3.11	0	0	8	2	2	1	13
3.12	0	2	7	1	4	1	15
3.13	1	2	1	2	1	9	16
3.14	0	0	1	1	9	2	13
3.15	0	0	2	1	7	1	11
3.16	2	2	2	2	3	8	19
3.17	1	1	1	0	1	5	9
3.18	0	0	0	0	0	2	2
3.0	**40**	**40**	**40**	**40**	**40**	**32**	**232**
4.1	1	1	1	2	10	1	16

ID	DL/CV	ML	NLP	RDL	FQA	PMI	Total
4.2	4	10	0	1	1	0	16
4.3	0	2	9	1	3	0	15
4.4	1	0	1	8	3	1	14
4.5	2	1	1	1	1	6	12
4.6	8	2	1	2	0	0	13
4.7	2	2	2	2	1	0	9
4.8	1	1	2	1	0	5	10
4.9	1	1	2	1	1	3	9
4.10	0	0	1	1	0	2	4
4.0	**20**	**20**	**20**	**20**	**20**	**18**	**118**
5.1	4	4	4	4	4	8	28
5.2	2	2	2	2	2	6	16
5.3	2	2	2	2	2	4	14
5.4	0	0	0	0	0	6	6
5.0	**8**	**8**	**8**	**8**	**8**	**24**	**64**
All	**98**	**98**	**98**	**98**	**98**	**98**	**588**

7.5 Working agreements and cover rule

1. Every role presents its own module in the final presentation (package 4.9), so visibility of the work is shared equally.
2. A review request is answered within one working day; a commit is pushed at least once per working day.
3. Each role writes a one-page handover note for its module by IM-2 (20 Nov). If a role is unavailable, its peer buddy covers the critical-path package, and the PM and Integration Lead moves hours between roles through a logged change request.
4. Hours are tracked weekly against this plan; a role that falls more than 4 hours behind raises it at the next sync instead of waiting for a gate.

8. Required Resources and Technologies

The project needs six people at 98 hours each, one public dataset, a Python machine-learning and web stack, and a shared repository; computing capacity is the one resource that is not yet verified.

All technologies below are **proposed** by the planning documents [2][3] or are standard choices for them. Exact versions are pinned in week 1 after the hardware check.

8.1 Human resources and skills

Role	Planned hours	Skills the role needs
DL/CV	98	PyTorch, transfer learning, image preprocessing, model evaluation
ML	98	pandas, scikit-learn, data splitting, calibration, evaluation metrics
NLP	98	Regular expressions, Unicode and Arabic text handling, fuzzy matching, careful annotation
RDL	98	Rule-table design, Python, test-case design for decision logic
FQA	98	Streamlit, pytest, test design, usability walkthroughs
PMI	98	FastAPI, Pydantic, Git, planning and risk tracking, technical writing

Any skill gap found in week 1 is covered by the peer buddy and by short self-study inside the role's own hours; no external staff is assumed.

8.2 Data resources

Resource	Use	Owner	Note
Fashion Product Images (Small), Kaggle [6]	Sole MVP dataset: images plus styles.csv linked by product ID; about 44,000 products according to the source listing	ML	Fields used: id, articleType, baseColour, productDisplayName, masterCategory, subCategory. Row counts are source documentation until EDA confirms them
Synthetic mismatch cases	Training and evaluation of the mismatch classifier and rules	ML	Generated only after the split; every record carries a provenance flag
Manual evaluation set	Independent English and Arabic listings built from test-split images	NLP, checked by FQA	Planned size 100 to 150, proposed target 120; locked read-only at DG-3
Vocabulary and color-family tables	One shared configuration used by CV, NLP and rules	RDL	Frozen at DG-2

No other dataset is used, and no real marketplace data is assumed.

8.3 Software and technologies

Area	Technology	Purpose	WBS
Language and notebooks	Python 3, Jupyter, virtual environment with pinned requirements file	Common environment for all roles	2.9, 3.17
Image models	PyTorch with torchvision ResNet or MobileNet pretrained weights; open-source CLIP implementation [10][11][12]	Transfer learning for type and color; zero-shot baseline and fallback	3.2, 3.3
Image processing	Pillow, OpenCV	Resolution, blur and brightness checks; dominant-color cross-check	3.4
Data and ML	pandas, NumPy, scikit-learn, Matplotlib	Splits, error injection, logistic regression or small MLP, metrics, plots	3.1, 3.7, 3.8, 4.2
NLP	Python regular expressions, Unicode script check for Arabic characters, a fuzzy-matching library	English rules, fixed Arabic lexicon, controlled vocabulary matching	3.6, 3.11
Backend	FastAPI, Uvicorn, Pydantic [13]	Orchestration of the CV and NLP branches and the shared data contract	3.13
Frontend	Streamlit [14]	Intake form, results, Review Queue, Before/After, seller decision	3.14, 3.15
Testing	pytest with fixture files	Unit tests, contract tests and AC-001 to AC-025	4.1
Collaboration	Git repository with pull requests, shared task board, shared folder, team chat	Code and document control, tracking, communication	5.1 to 5.4
Optional, first to be cut	CLIP linear probe, MLflow experiment logs, Docker package	Supporting reproducibility only if the core is secure	3.17

8.4 Computing and infrastructure

The plan needs a computer that can train a small image model on a subset of the dataset and run inference for one listing at a time, plus storage for the images and saved weights. Hardware capability, memory, GPU availability and access to pretrained-weight downloads are **not verified**; the hardware inventory is completed by 23 Oct (package 2.5). To keep training feasible the plan uses a lightweight backbone, reduced image sizes and frozen early layers. If local machines are too slow, shared cloud notebook services are the contingency, subject to availability (risk R-13).

No budget, cost estimate or return on investment is asserted; the stack is open source and the dataset is publicly listed. Real cost depends on local machines, team availability and licence terms.

8.5 Repository layout and version rules

```
smartlisting-copilot/
  data/         images and styles.csv stay local; split ID lists are committed
  notebooks/    EDA and experiments, named by WBS ID
  src/
    cv/  nlp/  rules/  ml/  api/  ui/
```

```
config/       vocabulary, color families, thresholds (frozen at DG-3)
tests/        unit tests, contract tests, AC fixtures
docs/         project documents with version numbers
requirements.txt  README.md
```

- The main branch is protected; every change arrives by a pull request reviewed by the peer buddy.
- Branches are named by WBS ID, for example feature/3.3-resnet-training.
- Each gate and milestone is tagged in the repository, for example gate-DG-3 and milestone-M2.
- Document files carry the project name, document name and version number.

9. Weekly Plan and Deliverables

Each of the seven weeks has one focus, named deliverables and a check; by the end of the week each deliverable is in the repository or the shared folder.

9.1 Weekly overview

Week	Dates	Focus	Weekly deliverables	Gate or check
1	19 to 23 Oct	Data audit, early design, project setup	Roles assigned and availability confirmed; repository, board and conventions ready; hardware inventory; EDA draft; CV and NLP design drafts; data contract v0.1 (23 Oct)	Week-1 checkpoint, Fri 23 Oct
2	26 to 30 Oct	Freeze classes and design inputs	Five classes frozen (28 Oct); split protocol (28 Oct); EDA report v1; vocabulary and color-family tables v1 (30 Oct); CV and NLP design v1	DG-1, Wed 28 Oct
3	2 to 6 Nov	Complete the design and submit	Data contract v1.0 (2 Nov); comparison and decision-logic design; error-injection and classifier design; UI wireframes and test fixtures; evaluation protocol; open decisions recorded (4 Nov); Analysis and Design document (freeze 5 Nov, submit 6 Nov)	DG-2, Mon 2 Nov; M1, Fri 6 Nov
4	9 to 13 Nov	Data pipeline and first modules	Splits and loaders (12 Nov); CLIP baseline; model training started (12 Nov); first versions of the NLP module and comparison engine; API and interface skeleton on stubs	IM-1, Fri 13 Nov
5	16 to 20 Nov	Real modules and integration	NLP module and comparison engine complete (18 Nov); synthetic error dataset (18 Nov); intake and results interface complete (18 Nov); image-quality checks; template generator; first real end-to-end run	IM-2 and scope-cut checkpoint, Fri 20 Nov
6	23 to 27 Nov	Final models, frozen thresholds, code freeze	Final CV model (23 Nov); mismatch classifier (24 Nov); Review Queue routing; Before/After and seller-decision interface (25 Nov); manual set complete (25 Nov); thresholds frozen (25 Nov); acceptance tests under way from 23 Nov; held-out evaluation starts 26 Nov; reproducibility package	DG-3, Wed 25 Nov; IM-3, Fri 27 Nov
7	30 Nov to 4 Dec	Submit, evaluate, report, present	Implementation submitted (30 Nov); evaluations complete (1 Dec); error analysis and defect fixes complete (2 Dec); final report and presentation drafted (2 Dec); two rehearsals (2 and 3 Dec); submission package check (3 Dec)	M2, Mon 30 Nov; internal ready Thu 3 Dec; M3, Fri 4 Dec

9.2 Weekly work by role

Entries show WBS package IDs and the work for that week.

Week	DL/CV	ML	NLP	RDL	FQA	PMI
1	2.1 image audit; 2.5 CV design start; hardware inventory	2.1 lead EDA	2.1 title audit; 2.6 NLP design start	2.1 color distribution; 2.3 vocabulary start	2.1 file and ID integrity checks; test tooling	2.9 architecture and contract v0.1; 5.4 repository and board
2	2.5 CV design v1; 2.2 class evidence	2.2 DG-1; 2.4 split protocol	2.6 NLP design v1; 2.3 aliases	2.3 groups and color families v1	2.10 wireframes start; 2.4 duplicate checks	2.9 contract draft; DG-1 facilitation; weekly status

Week	DL/CV	ML	NLP	RDL	FQA	PMI
3	Review 2.12 sections; 2.11 support	2.8 classifier design; 2.11 metrics	2.8 injection text variants; 2.7 support	2.7 decision-logic design; Q-01 and Q-02 proposals	2.10 fixtures; 2.11 protocol	2.9 DG-2; 2.12 consolidation and submission
4	3.2 CLIP baseline; 3.3 training start	3.1 data pipeline; 3.7 start	3.6 NLP module	3.9 comparison engine	3.14 intake interface	3.13 API and stubs; 3.16 integration plan
5	3.3 training; 3.4 quality checks	3.7 injection; 3.8 classifier start; 3.5 start	3.6 complete; 3.11 templates; 3.12 manual set	3.9 complete; 3.10 routing	3.14 complete; 3.15 start; 3.12 label check	3.16 real-module integration; IM-2 and cut checkpoint
6	3.3 final model; 3.5 support; 4.1 support	3.8 classifier; 3.5 DG-3; 4.2 start	3.11 and 3.12 complete; 4.3 start	3.10 complete; 4.4 scenario tests	3.15 complete; 4.1 acceptance tests	3.16 IM-3; 3.17 package; 3.18 freeze; 4.8 outline
7	4.6 error analysis; 4.7 fixes	4.2 complete; 4.7 fixes	4.3 complete; 4.7 fixes	4.4 complete; 4.7 fixes	4.1 sign-off; defect triage	3.18 submission; 4.5; 4.8; 4.9; 4.10

9.3 Weekly routine

1. **Monday:** weekly sync, risk review and plan for the week.
2. **Wednesday:** written status, one Green, Amber or Red line per owned package.
3. **Thursday:** review window; peer buddies finish pending reviews.
4. **Friday:** checkpoint; owners move cards, the PM and Integration Lead issues the one-page status, and any gate result is logged.

In weeks that end on an official date or an internal freeze, the review window moves one working day earlier.

10. Risk Management Plan

Risks are identified, scored, owned and reviewed every week; the baseline register holds 32 risks, four rated Severe and twelve rated High.

The six risks named in the brief are the first six register entries: Class Imbalance (R-01), Incorrect Labels (R-02), Color Ambiguity (R-03), Model Overfitting (R-04), Synthetic Data Bias (R-05) and Integration Issues (R-06). The other 26 come from the scope documents, the open decisions, the schedule and the team plan.

10.1 Process

The six steps below follow the stages of the ISO 31000:2018 risk-management process (identify, analyze, evaluate, treat, monitor and review, record and report) [9], reduced to the size of a seven-week team project. The five-point scales and bands in section 10.2 are the team's own planning judgments.

1. **Identify.** Risks come from the planning and requirements documents, from the EDA results, and from any role at any time.
2. **Assess.** Likelihood and impact are scored from 1 to 5 using the scales below; score is likelihood times impact.
3. **Plan the response.** Each risk has preventive actions, an early-warning trigger and a contingency (section 12).
4. **Monitor.** The owner watches the trigger; the PM and Integration Lead reviews the whole register at every weekly sync.
5. **Respond.** When a trigger is met the owner starts the contingency within one working day and logs it.
6. **Close.** A risk is closed when its exposure window has passed or its trigger can no longer occur; closed risks stay in the register.

10.2 Scoring scales

Probability bands are planning judgments, not statistics.

Score	Likelihood	Impact
1	Rare (below about 10%)	Negligible: absorbed inside one package, no deliverable affected
2	Unlikely (about 10 to 30%)	Minor: one to two days of slip or a small quality loss, absorbed by the buddy or the buffer
3	Possible (about 30 to 50%)	Moderate: a package or an internal gate is late or must be redone
4	Likely (about 50 to 80%)	Major: an official milestone or the quality of the core pipeline (a success criterion) is threatened
5	Almost certain (above about 80%)	Critical: a missed official date or an invalid evaluation

Rating	Score (likelihood times impact)	Management rule
Low	1 to 4	Review weekly; no special action

Rating	Score (likelihood times impact)	Management rule
Moderate	5 to 9	Owner tracks the trigger; contingency prepared
High	10 to 15	Preventive actions scheduled in the plan; discussed at every sync
Severe	16 to 25	Preventive actions start first; also checked in the Wednesday status; reported to the supervisor before each milestone if applicable

10.3 Responsibilities

- **Risk owner:** a technical role named in the register; keeps the score current, watches the trigger, runs the contingency.
- **PM and Integration Lead:** maintains the register (package 5.2), runs the weekly review, decides with the owner when a contingency or the cut order is activated.
- **Whole team:** raises new risks at the sync and takes part in gate decisions.

10.4 Baseline exposure

Rating	Count	Risks
Severe	4	R-03, R-05, R-06, R-14
High	12	R-01, R-02, R-04, R-07, R-08, R-10, R-11, R-12, R-13, R-15, R-17, R-30
Moderate	15	R-09, R-16, R-18, R-19, R-20, R-21, R-22, R-23, R-24, R-25, R-26, R-27, R-29, R-31, R-32
Low	1	R-28

The two most exposed areas are data and attribute quality (color ambiguity, synthetic data bias) and delivery (integration, schedule compression). Both receive preventive work in the first three weeks, before implementation starts.

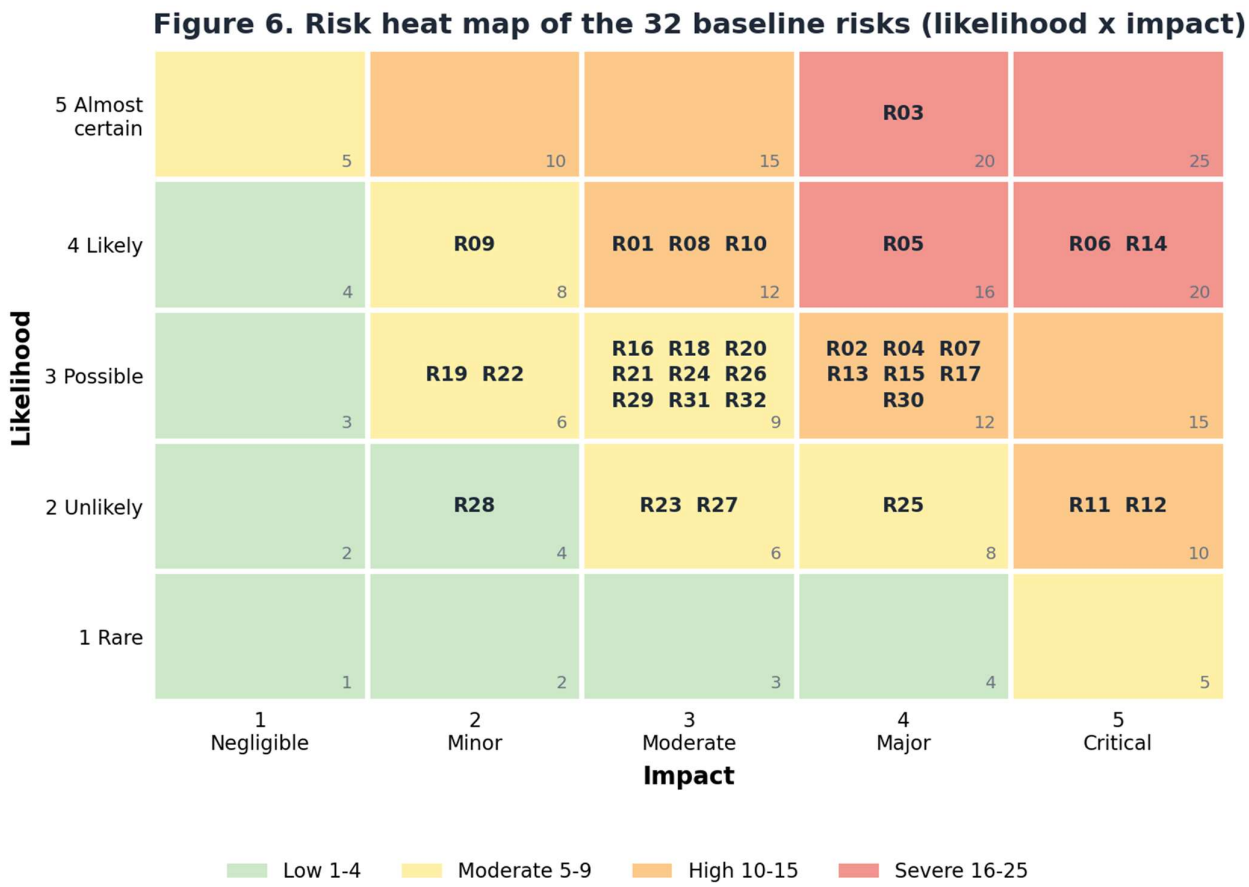


Figure 6 places each risk in the cell given by its likelihood (rows) and impact (columns). Cell colour follows the rating bands of section 10.2. Severe risks (R-03, R-05, R-06, R-14) are in the top-right, and 16 of the 32 risks are High or Severe. Colours carry no extra meaning beyond the rating; the number in the corner of each cell is the score.

11. Risk Register

All 32 risks are open at the baseline of 15 October 2026; each has one owner role and a score of likelihood (L) times impact (I) using the scales in section 10.2.

ID	Risk	Category	Cause and effect	L	I	Score	Rating	Owner	WBS
R-01	Class imbalance	Data	The chosen article types have very unequal image counts; the model favors large classes and rare-class recall and macro-F1 fall	4	3	12	High	ML	2.1, 2.2, 3.3
R-02	Incorrect or noisy labels	Data	articleType or baseColour labels in the metadata are wrong or inconsistent; training learns the errors and evaluation misstates accuracy	3	4	12	High	ML	2.1, 3.1
R-03	Color ambiguity	Data and logic	Multi-color items, near shades, lighting and unclear baseColour make a different color hard to define; false COLOR_MISMATCH flags or missed mismatches	5	4	20	Severe	RDL	2.3, 3.4, 3.9
R-04	Model overfitting	Model	A large model memorizes the training images of only five classes; a wide train-to-validation gap and a weak held-out result	3	4	12	High	DL/CV	3.3, 3.5
R-05	Synthetic data bias	Evaluation	Error cases are generated from templates; the classifier learns generator artifacts and metrics look better than real performance	4	4	16	Severe	ML	2.8, 3.7, 3.8, 4.2
R-06	Integration issues	Integration	CV, NLP, rules, API and interface are built in parallel; data structures, vocabularies or versions do not match when joined	4	5	20	Severe	PMI	2.9, 3.13, 3.16
R-07	Near-type visual confusion	Model	Similar product types in the five classes look alike at low resolution; type predictions are mixed up and TYPE_MISMATCH is raised wrongly	3	4	12	High	DL/CV	2.2, 3.3
R-08	Low resolution and background variation	Data	Small images and varied backgrounds hide detail; CV accuracy and the blur check become less reliable	4	3	12	High	DL/CV	3.3, 3.4
R-09	Limited Arabic lexicon coverage	NLP	The fixed Arabic vocabulary misses spellings, dialect words or stylized color names; Arabic claims are not extracted	4	2	8	Moderate	NLP	2.6, 3.6
R-10	English claim ambiguity	NLP	Titles with several colors, synonyms, or no explicit type or color give wrong or missing claims	4	3	12	High	NLP	3.6, 4.3
R-11	Split leakage and duplicate products	Evaluation	Images of one product, or injected variants of it, appear in both train and test; metrics are inflated	2	5	10	High	ML	2.4, 3.1, 3.7
R-12	Threshold tuning leakage	Evaluation	Thresholds are adjusted after seeing held-out or manual-set results; the final metrics are not valid	2	5	10	High	FQA	3.5, 4.2, 4.3
R-13	Compute and environment limits	Technical	No GPU, small memory or slow downloads make training exceed the time available	3	4	12	High	DL/CV	2.5, 3.3
R-14	Schedule compression	Schedule	Only four days separate code submission from the final presentation and seven weeks hold a multimodal build; packages slip into testing time	4	5	20	Severe	PMI	All, especially 3.16 and 4.x
R-15	Scope creep	Scope	Features outside the scope (price checks, LLM text, more attributes) are added; core work is delayed	3	4	12	High	PMI	5.4
R-16	Unresolved requirement decisions	Requirements	Q-01, Q-02 and IN-01 to IN-04 stay open; AC-020 and input handling cannot be signed off	3	3	9	Moderate	RDL	2.7, 3.10
R-17	Team availability and uneven contribution	Team	Exams, illness or other courses reduce a role's hours; its critical packages slip	3	4	12	High	PMI	5.1, all

ID	Risk	Category	Cause and effect	L	I	Score	Rating	Owner	WBS
R-18	Poor calibration and warning overload	Model and logic	Confidence values are unreliable or thresholds too loose; false alarms and uncertain listings fill the Review Queue	3	3	9	Moderate	RDL	3.5, 3.10
R-19	Mismatch classifier adds no gain over rules	Model	Synthetic training data give no improvement over the rule baseline; the classifier effort brings no benefit	3	2	6	Moderate	RDL	3.8, 4.2
R-20	CLIP fallback and unsupported-category detection unreliable	Model	CV and CLIP disagree too often or the other-product prompt fails; out-of-scope images are forced into a class or valid images are flagged	3	3	9	Moderate	DL/CV	3.2, 3.4
R-21	Dependency and reproducibility failures	Technical	Library versions, saved weights or file paths differ between machines; the code does not run for the assessors	3	3	9	Moderate	FQA	3.17, 3.18
R-22	Version-control conflicts and lost work	Technical	Merge conflicts, overwritten files or lost notebooks delay integration	3	2	6	Moderate	PMI	3.16, 5.4
R-23	Unclear explanations and usability	Usability	Reason codes, severity or Before/After wording is hard for a seller to understand	2	3	6	Moderate	FQA	2.10, 3.14, 3.15
R-24	Manual set under-powered or subjectively labeled	Evaluation	A set of 100 to 150 listings and subjective color labels give wide uncertainty; conclusions are weak	3	3	9	Moderate	NLP	3.12, 4.3
R-25	Demo failure	Delivery	The live demo fails because of environment, model loading time or network	2	4	8	Moderate	FQA	4.9
R-26	Documentation and traceability drift	Documentation	IDs, dates or scope statements differ between the six project documents; reviewers find contradictions	3	3	9	Moderate	PMI	4.8, 5.4
R-27	Inference too slow	Technical	One listing takes too long to analyze; response-time measurement and the demo suffer	2	3	6	Moderate	DL/CV	3.3, 4.5
R-28	Dataset terms and access	Data	Terms of use or download access differ from the assumption; data cannot be shared or reproduced	2	2	4	Low	ML	2.1, 3.17
R-29	Suggested fix repeats a wrong prediction	Logic and UX	A template writes the wrong type or color because the CV prediction was wrong; the seller is misled	3	3	9	Moderate	NLP	3.11, 4.4
R-30	Vocabulary mismatch between text claims and image labels	Integration	NLP normalizes to names that differ from CV label names; comparison fails or flags false mismatches	3	4	12	High	NLP	2.3, 3.6, 3.9
R-31	Rule-table brittleness	Decision logic	Type-group boundaries or near-shade tolerance are wrong; severity labels (Critical or Major) are wrong	3	3	9	Moderate	RDL	2.3, 3.9, 4.4
R-32	Acceptance fixtures incomplete or blocked	Testing	Fixtures are missing or blocked by open decisions; some of AC-001 to AC-025 cannot be signed off	3	3	9	Moderate	FQA	2.10, 4.1

12. Risk Mitigation Strategies

Every risk has preventive actions that are scheduled inside the plan, an early-warning trigger, and a contingency that keeps the approved scope intact.

12.1 Strategy for each risk

ID	Preventive actions	Early-warning trigger	Contingency	When
R-01	Choose the five classes with class counts in view (2.2); stratified product-ID splits; class-weighted loss or balanced sampling; report macro-F1 and per-class recall, not accuracy alone	One class is far smaller than the others (limit agreed at DG-1), or per-class validation recall is far below the average	Replace the smallest class before DG-1; augment minority-class images; undersample the largest class; document the limitation	19 to 28 Oct; monitor 12 to 25 Nov
R-02	Visual audit of a stratified sample of articleType and baseColour labels during EDA; written rule for dropping or correcting clearly wrong records; shared vocabulary (2.3)	The audit finds many wrong labels, or the same confusions recur across training epochs	Exclude noisy records and ambiguous classes; evaluate color on the confident-label subset and report both results	19 to 28 Oct
R-03	Color-family and near-shade table with aliases (2.3); policy for multi-color items; dominant-color cross-check (3.4); only different-family contradictions are Major; ambiguous cases go to Needs review; near shades, missing claims and multi-color cases included in the manual set	Neighboring color families are often confused on validation data, or CV color and dominant color often disagree	Widen the near-shade tolerance before DG-3; route ambiguous colors to Needs review instead of Major; restrict color evaluation to clearly separable families and say so	22 to 30 Oct; tolerance final at DG-3 (25 Nov)
R-04	Frozen backbone first, then fine-tune a few layers; augmentation, weight decay or dropout, early stopping on validation loss; track the train-to-validation gap	Validation loss rises while training loss falls, or the gap grows over epochs	Freeze more layers, stronger augmentation, or a smaller MobileNet; use CLIP zero-shot as the type baseline	12 to 23 Nov
R-05	Split by product ID before injection (2.4); several injection strategies with varied text (real titles, rewrites, paraphrases); evaluate the classifier on the manual set separately; keep rules authoritative; provenance flag on every record	Near-perfect synthetic score but weak manual-set score, or an ablation shows the classifier relies on template tokens	Base decisions on rules; regenerate the data with more varied text; report the gap openly	29 Oct to 4 Nov; 12 to 24 Nov
R-06	Data contract v1.0 frozen at DG-2; stubs by IM-1; contract tests in the repository; each owner publishes a sample output; integration build every Friday from week 4; one shared vocabulary file	A contract test fails, IM-1 or IM-2 is missed, or two modules disagree on a field	Revert the blocked module to its stub; daily integration pair sessions in weeks 5 and 6; apply the cut order	20 Oct to 27 Nov
R-07	Select visually separable classes (2.2); run a quick CLIP baseline to expose confusing pairs; group-level severity makes same-group confusion Major, not Critical	The confusion matrix shows a pair that is often mixed up	Replace one class of the pair before DG-1; send low-confidence cases to Needs review	26 to 28 Oct
R-08	Model-specific preprocessing with no claim of recovering hidden detail; blur and quality flags route to Needs review; prefer classes that stay recognizable at low resolution	Validation error review shows accuracy dropping on small or cluttered images	Standard upscaling; route low-quality images to Needs review; state the limitation	16 to 25 Nov

ID	Preventive actions	Early-warning trigger	Contingency	When
R-09	Build the fixed Arabic lexicon from the five classes and the color families with common spelling variants; Arabic cases in the manual set; keep scope bounded	Low recall on the Arabic part of the manual set	Add terms inside the bounded scope before DG-3; an unknown Arabic term yields no claim, so a missing-attribute flag or Needs review, never a false mismatch	20 to 30 Oct; 9 to 18 Nov
R-10	Controlled vocabulary with synonyms and fuzzy matching; rule for titles with several colors; unit tests with real dataset titles	Low recall on validation titles	Treat ambiguous claims as unknown; add synonyms; keep ambiguous text out of mismatch decisions	9 to 18 Nov
R-11	Product-ID split first; duplicate and image-hash checks across splits; automated leakage test run by FQA before any training	Any duplicate found across splits	Rebuild the splits and retrain the affected models, ideally before 12 Nov	26 to 28 Oct; check on 12 Nov
R-12	Thresholds kept in one configuration file frozen at DG-3; manual set locked read-only; FQA audits changes to both after DG-3	Any change to thresholds or the manual set after 25 Nov	Disclose in the report; re-run on an untouched subset if one exists, otherwise report as a limitation	From 25 Nov
R-13	Hardware inventory in week 1; lightweight backbone, reduced image size, frozen early layers; cache preprocessed images; training time per epoch estimated	First test run exceeds the planned time or memory	Switch to MobileNet or smaller images; train on a class-balanced subset; use shared cloud notebooks if available; CLIP baseline as fallback	19 to 23 Oct; 12 Nov
R-14	Internal freezes one working day before official dates; critical-path tracking; stub-first integration; testing overlapping from 23 Nov; at most two active packages per person; cut order	A critical-path package is more than one working day late, or IM-1 or IM-2 is missed	Apply the cut order; the buddy takes over part of the package; stop all new features	Whole project; checks at each IM gate
R-15	Written out-of-scope list; change requests with impact; default answer is no; supervisor consulted on scope	A task or request appears without a WBS ID	Park the request in a post-project list; accept no change after DG-2 unless it removes work	Continuous
R-16	Decision register with owners; proposals for Q-01 and Q-02 drafted by RDL by 2 Nov; decisions recorded by 4 Nov; routing table kept in a configuration file so either choice can be applied without rewriting code	A decision is still open on 4 Nov	The team records its own decision with evidence and date and marks it pending supervisor confirmation; AC-020 edge cases stay flagged as partially specified	26 Oct to 4 Nov
R-17	Equal 98-hour plan; availability and exam calendar collected in week 1; peer buddies; handover notes by IM-2; weekly hours tracking	A role is more than 4 hours behind plan, or misses a sync	The buddy covers the critical package; the PM rebalances hours by change request; apply the cut order	Week 1; continuous
R-18	Calibration check (ECE) on validation data (3.5) [15]; thresholds chosen on validation; false-alarm rate tracked; severity logic keeps minor cases out of Critical	Most test listings land in Needs review, or ECE is high	Re-calibrate (for example temperature scaling); adjust the confidence threshold before DG-3; report the false-alarm rate	20 to 25 Nov
R-19	Rules stay the authoritative baseline; classifier compared with rules on validation data; expectation set in the design	No improvement over rules on validation data	Keep the classifier as a documented supporting signal and report the negative result; decisions stay rule-based	18 to 24 Nov
R-20	Prompt set includes an other-product prompt; the disagreement rule is tested on mixed images including out-of-scope items; no forced definitive type	Many valid images are flagged, or out-of-scope images are accepted	Tighten or loosen the trigger before DG-3; send disputed cases to Needs review	9 to 20 Nov

ID	Preventive actions	Early-warning trigger	Contingency	When
R-21	Pinned requirements file; clean-environment run by FQA at IM-2 and IM-3; fixed random seeds; saved weights and README	A clean-environment run fails	Docker package if time allows; otherwise exact setup steps and an environment export	20 and 27 Nov
R-22	Protected main branch; pull requests with buddy review; push daily; gate tags; large files kept outside Git	A merge conflict stays unresolved for more than one working day	Restore from the last tag; the PM resolves it with both authors	Continuous
R-23	Plain-language explanation templates; wireframes reviewed in week 3; scenario walkthrough at IM-2 with a teammate who did not build the screen	The reviewer cannot explain what a screen means	Simplify the wording and reduce the fields shown	26 Oct to 4 Nov; 20 Nov
R-24	Target 120 listings within the 100 to 150 range; labeling guide; a second person checks every label; cover near shades, missing claims, swapped descriptions, multiple colors and stylized language	Author and checker disagree often, or the set is below the minimum on 22 Nov	Report the uncertainty and limits; extend the set if time allows; never merge manual and synthetic results	13 to 25 Nov
R-25	Demo script; two rehearsals (2 and 3 Dec); pre-loaded sample listings; weights stored locally; backup recording	A rehearsal fails	Present the recording and screenshots	30 Nov to 3 Dec
R-26	One ID scheme (OBJ, FR, AC, SC, WBS, R, DG, IM); traceability check by the PM at every gate; version number on each document; consistency pass before submission	An ID or date differs between documents	Reconcile with a change-log entry before submission	Each gate; 2 to 6 Nov; 3 Dec
R-27	Lightweight model; single-listing inference benchmark at IM-2; load models once at start-up	Slow response at IM-2	Smaller model or input size; caching	20 Nov; 30 Nov to 1 Dec
R-28	Read the dataset terms in week 1 (2.1); cite the source; do not redistribute images; commit only ID lists and code	Terms or access differ from the assumption	Ask the supervisor; share code and split ID lists only	19 to 23 Oct
R-29	Suggestions only from supported type and color; no definitive suggestion when LOW_CONFIDENCE or UNSUPPORTED_CATEGORY applies, only Please verify; the seller always decides; groundedness check (SC-06)	A scenario test shows a suggestion for an uncertain case	Switch the suggestion off for uncertain cases	16 to 24 Nov
R-30	One shared vocabulary file of canonical type and color names and aliases (2.3) used by CV, NLP and rules; mapping unit tests; the contract carries canonical names only	An unmapped label appears in a contract test	Add a mapping adapter; fix the vocabulary file and re-run	22 to 30 Oct; 9 to 18 Nov
R-31	Type groups derived from the dataset hierarchy (masterCategory, subCategory) after EDA and reviewed; cross-group and same-group fixtures (AC-010, AC-017)	A fixture gives a different severity than expected	Adjust the group table before DG-3; keep the tables in configuration	26 Oct to 4 Nov; 16 to 23 Nov
R-32	Fixtures designed in 2.10, one set per AC; FQA tracks coverage on the board; blocked ACs identified at M1	An AC has no fixture by IM-2	Prioritize ACs for the core pipeline; document blocked ACs (AC-020 edge cases) openly	26 Oct to 4 Nov; 20 Nov

12.2 Response principles

1. **Prevent early.** Most preventive actions fall in weeks 1 to 3, before implementation, because data and design risks are cheapest to fix then.
2. **Trigger, then act.** A contingency starts only when its trigger is met, within one working day, and is written in the change log.
3. **Keep the scope.** No contingency adds an out-of-scope feature or relaxes the evaluation rules in section 2.8.
4. **Protect the core.** When time is short, supporting features go first (section 12.4); the core pipeline never does.

12.3 Cross-cutting safeguards

- **Internal freezes** one working day before 6 Nov, 30 Nov and 4 Dec (R-14, R-25, R-26).
- **Stub-first integration** with gates IM-1 to IM-3 (R-06, R-14, R-30).
- **Split-before-injection and frozen thresholds** (R-05, R-11, R-12).
- **Configuration-driven rules** for groups, near shades and routing, so a decision can change without rewriting code (R-03, R-16, R-31).
- **Peer-buddy cover** for any unavailable role (R-17, R-22).
- **Bug-fix-only window** from 27 Nov to 2 Dec (R-14).

12.4 Scope fallback and cut order

The approved fallback order comes from the Revised Scope as recorded in [3]. Supporting features are cut in this order; reason codes, the Review Queue, suggested fixes and seller control are never cut.

Cut	Item	Requirement	When it is applied
1	Listing Quality Score	FR-018	At IM-2 (20 Nov) if any core package is Red, or at once if a trigger for R-06 or R-14 is met
2	Missing Attribute Detection	FR-012	At IM-3 (27 Nov) if a core package is still Red, or after cut 1 if the delay continues
3	MLflow experiment logs	Optional	Not started until the core is secure; dropped at IM-2 if the core is late
4	Docker package	Optional	Not started until the core is secure; dropped at IM-2 if the core is late

The optional CLIP linear probe is likewise only started when the core is secure. If FR-018 is cut, FR-006 still reports image-quality findings and review flags; the numeric score and its test AC-018 are then not claimed [3].

13. Critical Path, Dependencies and Traceability

The critical path runs from the data audit through class freeze, splits, model training and threshold freeze to the held-out evaluation and the final report, and it has only about one working day of buffer.

13.1 Critical path

Step	WBS	Dates	Why it cannot slip
1	2.1 Data audit and EDA	19 to 23 Oct	Class choice and label decisions depend on it
2	2.2 Five-class freeze (DG-1)	26 to 28 Oct	Splits, vocabulary and models all depend on the class list
3	2.4 Split protocol	26 to 28 Oct	Splitting must come before error injection
4	3.1 Data pipeline and splits	9 to 12 Nov	Training, calibration and the manual set need the splits
5	3.3 Transfer-learning training	12 to 23 Nov	Longest package on the critical path (8 working days); calibration needs the final model
6	3.5 Calibration and thresholds (DG-3)	20 to 25 Nov	Thresholds must be frozen before any held-out result
7	4.2 Held-out evaluation	26 Nov to 1 Dec	Source of the main evaluation evidence
8	4.6 Confidence and error analysis	1 to 2 Dec	Needed for the limitations in the report
9	4.8 Final report	25 Nov to 3 Dec	Must be ready one working day before submission
10	4.10 Submission package check	3 to 4 Dec	Official date 4 Dec

Three near-critical chains also need weekly attention: the integration chain (2.9, 3.13, 3.16, 4.1, 4.7), the text-to-decision chain (2.3, 3.6, 3.9, 3.10, 4.4) and the mismatch-data chain (2.4, 3.7, 3.8, 4.2).

13.2 Key dependencies

Predecessor	Successor	Why
2.1 EDA	2.2 Class freeze	Class balance, image availability and label quality come from EDA
2.2 Class freeze (DG-1)	2.3, 2.4, 3.1	Groups, splits and the pipeline are built for the frozen class list
2.3 Vocabulary and color families	2.8, 3.6, 3.9	CV, NLP, injection and rules must use the same names
2.4 Split protocol, then 3.1	3.7 Error injection	Injected variants of one product must not cross split boundaries
2.9 Data contract (DG-2)	3.13 API, 3.14 interface	Interface and backend are built against one contract
3.1 Splits	3.3, 3.5, 3.7, 3.12	Training, calibration, injection and the manual set use the splits

Predecessor	Successor	Why
3.3 Trained model	3.5 Calibration	Calibration needs the final model outputs
3.5 Thresholds (DG-3)	4.2, 4.3	Thresholds are frozen before held-out evaluation
3.9 Comparison engine	3.10 Queue routing	Routing uses severity and reason codes
3.11 Templates	3.15 Before/After	The interface shows the template output
3.12 Manual set	4.3 Manual-set evaluation	The set must be complete and locked first
3.16 Integration (IM-3)	4.1 Acceptance tests	Full tests need the integrated system
4.2, 4.3, 4.6	4.8 Final report	Results and limitations feed the report

13.3 Objectives to packages, requirements and success criteria

Objective	Main WBS packages	Requirements	Success criteria	Lead roles
OBJ-01 End-to-end pipeline	2.9, 3.13, 3.14, 3.16, 4.1	FR-001 to FR-003 and integration of all	SC-01	PMI, FQA
OBJ-02 Visual Type and Color	2.1, 2.2, 2.5, 3.1 to 3.5, 4.2, 4.6	FR-004 to FR-006, FR-013	SC-02	DL/CV, ML
OBJ-03 Text claims	2.3, 2.6, 3.6, 3.12, 4.3	FR-007, FR-008	SC-09	NLP
OBJ-04 Discrepancy detection	2.3, 2.7, 2.8, 3.7 to 3.9, 4.2	FR-009 to FR-011, FR-017	SC-03	RDL, ML
OBJ-05 Evidence and reason codes	2.7, 3.9, 3.14	FR-013, FR-016, FR-017	SC-04	RDL, NLP
OBJ-06 Review prioritization	2.7, 3.4, 3.10, 4.4	FR-006, FR-012, FR-014, FR-015, FR-019, FR-020	SC-05, SC-08	RDL
OBJ-07 Suggested corrections	3.11, 3.15	FR-021, FR-022	SC-06	NLP
OBJ-08 Seller control	2.10, 3.15, 4.1	FR-023 to FR-025	SC-07	FQA
OBJ-09 Transparent evaluation	2.4, 2.11, 3.5, 3.7, 3.12, 3.17, 4.2, 4.3, 4.5, 4.6, 4.8	None (evaluation deliverable)	SC-02, SC-03, SC-09, SC-10	ML, PMI

13.4 Success criteria to verification packages

Criterion	Verified in	Evidence
SC-01 End-to-end submission	3.16, 4.1	Scenario run from submission to seller decision
SC-02 Attribute performance	4.2, 4.6	Accuracy, macro-F1 and class and color breakdowns reported separately
SC-03 Contradiction identification	4.2	Precision, recall, F1 and false alarms for wrong-type, wrong-color, combined and correct cases
SC-04 Explainability	4.1	Every displayed warning shows a reason code, evidence and rationale

Criterion	Verified in	Evidence
SC-05 Review routing	4.4	Queue results match the agreed decision logic in test scenarios
SC-06 Suggested corrections	4.1, 4.4	Templates contain only supported type and color; Before/After highlights the change
SC-07 Human control	4.1	Accept, Edit and Ignore work; nothing is applied or published automatically
SC-08 Uncertain cases	4.4	LOW_CONFIDENCE, IMAGE_BLUR and UNSUPPORTED_CATEGORY cases are escalated
SC-09 Evaluation separation	4.2, 4.3	Synthetic and manual results reported separately
SC-10 Complete pipeline report	4.5, 4.8	Catch rate, response time, thresholds, provenance and limitations documented

13.5 Functional requirement coverage

All 25 requirements in [3] appear in at least one work package.

Requirements	Design package	Build package	Test package
FR-001 to FR-003	2.10	3.14	4.1
FR-004 to FR-006	2.5	3.2 to 3.4	4.1, 4.2
FR-007, FR-008	2.6	3.6	4.1, 4.3
FR-009 to FR-012	2.7, 2.8	3.8, 3.9	4.1, 4.2
FR-013, FR-014	2.7	3.3, 3.5, 3.10	4.1, 4.4
FR-015	2.5, 2.7	3.4, 3.10	4.4
FR-016, FR-017	2.7	3.9	4.1, 4.4
FR-018 to FR-020	2.7	3.10	4.1, 4.4
FR-021, FR-022	2.10	3.11, 3.15	4.1
FR-023 to FR-025	2.10	3.15	4.1

14. Assumptions, Open Decisions and Conclusion

Eight assumptions and fourteen open decisions carry the plan; each has an owner and a date by which it is checked or decided.

14.1 Planning assumptions

ID	Assumption	Checked by	Owner	If it is false
A-01	Six people each fill one role and can give about 14 hours per week for seven weeks (98 hours)	Week-1 checkpoint, 23 Oct	PMI	Rebalance hours by change request; consider the cut order earlier
A-02	The team follows the DEPI dates in [4]: 16 Oct, 6 Nov, 30 Nov and 4 Dec	Consistent with the 16 Oct submissions; confirm with the supervisor if applicable	PMI	Re-baseline the plan within two working days
A-03	The dataset files and styles.csv join reliably by ID, and at least five article types have enough usable labels	EDA, 23 Oct	ML	Change the class selection before DG-1
A-04	Team hardware can train a small image model and run single-listing inference	Hardware inventory, 23 Oct	DL/CV	Lightweight backbone, cloud notebooks, CLIP fallback (R-13)
A-05	Pretrained model weights can be downloaded	23 Oct	DL/CV	Use another source or a local copy; CLIP fallback
A-06	The supervisor, if applicable, can answer open decisions in the week they are raised	Ask by 2 Nov	PMI	The team decides and marks the decision pending supervisor confirmation
A-07	The manual evaluation set of about 120 listings can be written and checked in the time planned	Progress check, 22 Nov	NLP	Report with a smaller set and state its limits (R-24)
A-08	A shared repository, task board and communication channel exist from week 1	23 Oct	PMI	Set up within two working days

14.2 Open decisions

These decisions come from the Functional Requirements Specification [3] and the planning documents [2]. Nothing listed here is treated as an accepted rule until the decision, its evidence and its date are recorded.

ID	Open decision	Owner	Decision due	Affects
Q-01	Review Queue state for exactly one Major mismatch (Medium inconsistency level)	RDL	Wed 4 Nov	FR-020, AC-020
Q-02	Precedence when a Critical or High finding occurs together with review-only flags	RDL	Wed 4 Nov	FR-020, AC-020
IN-01	Whether an empty Product Title is rejected or treated as absent claims	FQA with NLP	Wed 4 Nov	FR-002, AC-002
IN-02	Whether an empty Product Description is rejected or treated as missing claims	FQA with NLP	Wed 4 Nov	FR-002, FR-012, AC-012
IN-03	How unreadable, corrupt or unsupported image files are detected and reported	FQA with PMI	Wed 4 Nov	FR-001, AC-001

ID	Open decision	Owner	Decision due	Affects
IN-04	Handling of a missing or malformed seller category	FQA with RDL	Wed 4 Nov	FR-003, FR-015
D-01	The final five articleType classes	ML	Wed 28 Oct (DG-1)	FR-004, FR-015
D-02	Type-group boundaries and color-family table	RDL	Version 1 on Fri 30 Oct; frozen Mon 2 Nov (DG-2)	FR-010, FR-011, FR-017
D-03	Near-shade tolerance (ignored or Minor)	RDL	Wed 25 Nov (DG-3)	FR-011, FR-017
D-04	Confidence threshold, selected on validation data	ML	Wed 25 Nov (DG-3)	FR-013, FR-014
D-05	Blur thresholds for IMAGE_BLUR	DL/CV	Wed 25 Nov (DG-3)	FR-006
D-06	Score weights, only if FR-018 is retained	RDL	Fri 20 Nov (IM-2)	FR-018
D-07	How long and in what form seller decisions are kept in the proof of concept	FQA	Wed 4 Nov	FR-023 to FR-025
D-08	Size of the manual evaluation set within 100 to 150 (proposed 120)	NLP	Fri 6 Nov	OBJ-09, SC-09

14.3 Conclusion and first actions

This plan gives the team one schedule, one set of gates and one risk register that all lead to the four official dates, with six technical roles at equal workload. Its main exposures are color ambiguity, synthetic data bias, integration between parallel modules, and the four days between code submission and the final presentation. Each has preventive work in the first three weeks and a defined fallback that protects the core pipeline.

First actions for week 1:

1. Assign the six roles and confirm each person's 14 hours per week.
2. Set up the repository, task board and file conventions (package 5.4).
3. Download the dataset, read its terms and complete the hardware inventory.
4. Start the EDA and the CV, NLP and architecture designs.
5. Publish data contract v0.1 on 23 Oct.

15. References

[1] SmartListing Copilot, Project Proposal (Revised Scope: Fashion Product Images Small), 12 October 2026. Primary scope authority; its content is used as quoted in [2] and [3].

[2] SmartListing Copilot, Project Planning Document (Revised), 16 October 2026. Source of OBJ-01 to OBJ-09, SC-01 to SC-10, the dataset plan and the evaluation rules.

[3] SmartListing Copilot, Functional Requirements Specification (Revised), 16 October 2026. Source of FR-001 to FR-025, AC-001 to AC-025, the open decisions Q-01, Q-02 and IN-01 to IN-04, and the fallback cut order.

[4] Project Documentation Guidelines, official submission schedule (DEPI column), supplied as an image: 16 Oct, 6 Nov, 30 Nov and 4 Dec 2026.

[5] Project Management brief: required items WBS, Gantt Chart, Project Timeline, Milestones, Team Roles and Responsibilities, Risk Management Plan, Risk Mitigation Strategies, Required Resources and Technologies, Weekly Plan and Deliverables.

[6] P. Aggarwal, "Fashion Product Images (Small)," Kaggle dataset, 2019. kaggle.com/paramaggarwal/fashion-product-images-small (accessed October 2026).

[7] Project Management Institute, A Guide to the Project Management Body of Knowledge (PMBOK Guide) and The Standard for Project Management, 7th ed. Newtown Square, PA: PMI, 2021.

[8] K. Schwaber and J. Sutherland, The Scrum Guide. scrumguides.org, November 2020.

[9] International Organization for Standardization, ISO 31000:2018, Risk management: Guidelines. Geneva: ISO, 2018.

[10] A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," Advances in Neural Information Processing Systems 32 (NeurIPS), 2019. arXiv:1912.01703.

[11] A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," Proc. ICML (PMLR 139), 2021. arXiv:2103.00020.

[12] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov and L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," Proc. IEEE/CVF CVPR, 2018, pp. 4510-4520. arXiv:1801.04381.

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[15] C. Guo, G. Pleiss, Y. Sun and K. Q. Weinberger, "On Calibration of Modern Neural Networks," Proc. ICML (PMLR 70), 2017, pp. 1321-1330. arXiv:1706.04599.

Reference use. [1] to [3] define scope, objectives and requirements; [4] governs all dates; [5] governs the contents of this document; [6] identifies the data source; [7] to [9] support the management approach and the risk process; [10] to [14] identify the planned model and application frameworks; [15] supports the calibration check. All external sources [6] to [15] were published or last updated between 2017 and 2026, within the last ten years; [1] to [5] are project documents of October 2026. No interview, approval or measured result is cited or claimed in this document.